

GOVERNMENT OF KERALA
KERALA STATE PLANNING BOARD

FOURTEENTH FIVE-YEAR PLAN
(2022-2027)

WORKING GROUP ON ADDRESSING ISSUES RELATED TO HUMAN-WILDLIFE INTERACTIONS IN KERALA

REPORT

AGRICULTURE DIVISION
March 2022

FOREWORD

Kerala is the only State in India to formulate and implement Five-Year Plans. The Government of Kerala believes that the planning process is important for promoting economic growth and ensuring social justice in the State. A significant feature of the process of formulation of Plans in the State is its participatory and inclusive nature.

In September 2021, the State Planning Board initiated a programme of consultation and discussion for the formulation of the 14th Five-Year Plan. The State Planning Board constituted 44 Working Groups, with more than 1200 members in order to gain expert opinion on a range of socio-economic issues pertinent to this Plan. The members of the Working Groups represented a wide spectrum of society and include scholars, administrators, social and political activists and other experts. Members of the Working Groups contributed their specialised knowledge in different sectors, best practices in the field, issues of concern, and future strategies required in these sectors. The Report of each Working Group reflects the collective views of the members of the Group and the content of each Report will contribute to the formulation of the 14th Five-Year Plan. The Report has been finalised after several rounds of discussions and consultations held between September to December 2021.

This document is the Report of the Working Group on "Addressing issues related to human-wildlife interactions." The Co-Chairpersons of Working Group were Sri. P K Kesavan, IFS and Sri. Bennichan Thomas, IFS. Dr. R.Ramakumar, Member of the State Planning Board co-ordinated the activities of the Working Group. Sri.S.S.Nagesh, Chief, Agriculture Division was the Convenor of the Working Group and Dr. Reji D.Nair, Research Officer, Agriculture Division was Co-Convenor. The terms of reference of the Working Group and its members are in Appendix 1 of the Report

Member Secretary

PREFACE

As part of formulation of the 14th Five Year Plan, the Kerala State Planning Board had constituted working groups of experts in all the major sectors. In Agriculture and Allied Sectors, 6 working groups were constituted viz. Agriculture and Cooperation, Animal Husbandry and Dairy, Inland and Marine Fisheries, Forest and Environment, Water Resources and Regional Packages. To discuss and frame policies in each of these sectors, the working groups were further divided into 28 Expert Sub-Groups (ESG) with specific mandates.

Each Expert Subgroup held at least three meetings beside one focused group meeting before finalising the report. We, the Co-Chairs, place our deep appreciation and gratitude to all the esteemed members of the ESG for their valuable contributions in preparing the report. We are extremely grateful to Dr. V. K. Ramachandran, the Honourable Vice-Chairperson, Kerala State Planning Board, Dr. R. Ramakumar, Member, Kerala State Planning Board and Sri. S. S. Nagesh, Chief, Agriculture Division for their consistent guidance and suggestions in preparing the report. The drafting team put in commendable work in bringing together all the views and opinions of the members. We sincerely hope the recommendations in the report can lead to important changes in the public policy for the development of Forest and Wildlife sectors in the State.

Sri.Bennichan Thomas, IFS
Expert co-chairperson

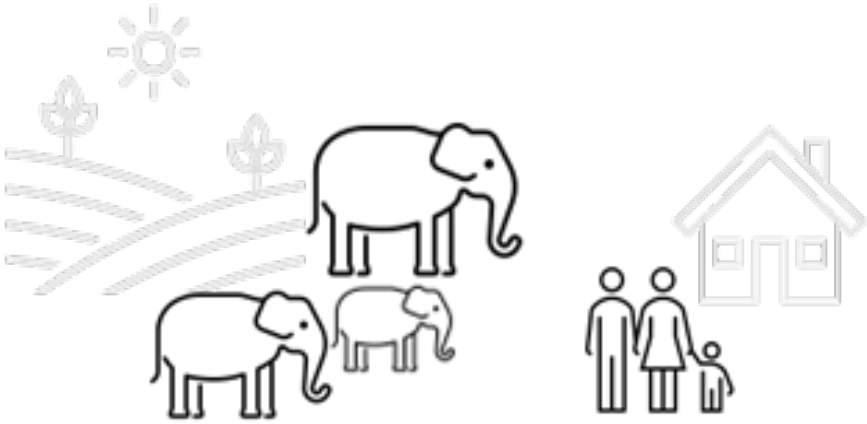
Sri. P K Kesavan, IFS
Official co-chairperson

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HIGHLIGHTS

- Mitigation measures for human-animal conflicts should be developed on the basic premise of meaningful engagement with and participation of all primary stakeholders, especially local communities.
- Human-wildlife conflict is a threat in almost all forest divisions of Kerala, especially in the northern region, but mitigation measures have been only partially effective.
- This report provides a comprehensive set of recommendations as well as guidelines for the allocation of physical and budgetary provisions.



ADDRESSING ISSUES RELATED TO HUMAN-WILDLIFE INTERACTIONS IN KERALA

EXECUTIVE SUMMARY

INTRODUCTION: Human-wildlife conflicts are largely a human-induced phenomenon, compounded by the species-specific behavioural ecology of animals and external environmental factors. Thus, mitigation measures for all negative interactions should be developed on the basic premise of meaningful engagement with and participation of all primary stakeholders, especially local communities.

These past few years have seen escalated tensions in wildlife management in Kerala, and settlers near reserve forests and sanctuaries express a heightened sense of insecurity. The growing pressure of the ever-increasing human population, habitat loss and fragmentation, changes in cropping patterns, and increasing wildlife population are considered as the major causes. A comprehensive wildlife management methodology that integrates the social and biological sciences and maximizes community participation is the need of the hour to reduce such conflict.

ANALYSIS OF PRESENT SITUATION: Changed cropping patterns, modification of the forest environment, and human incursions into traditional migration paths of wild animals result in crop-raiding and aggressive behaviour. The socioeconomic effects on the human population must also be considered while formulating a management strategy. Human-wildlife conflict is a threat in almost all forest divisions of Kerala, especially in the northern region. A more recent phenomenon is the proliferation of conflict issues in areas far removed from forest fringes. There are several methods used to prevent crop-raiding by wild animals, including trenches, solar-powered fences, translocation of animals, etc., and strategies to recompense losses by ex-gratia payments. However, these have only been partially successful, and an integrated strategy is called for.

RECOMMENDATIONS: This report recommends the following broad strategies to mitigate human-wildlife conflicts in the state.

1. Creation of physical barriers to prevent animals from reaching the human habitations and crop fields. The barriers will be used judiciously taking a landscape approach so as not to disrupt the natural movement of the animals which is essential to the conservation of the elephants in the landscape
2. Adoption of measures to avoid the interface between humans and wild animals.
3. Developing infrastructure and human resources for effective mitigation of human-wildlife conflicts.
4. Providing ex-gratia, crop insurance, life insurance, and ex-gratia to the victims of wildlife damage.
5. Involving Panchayaths and line departments to assist in conflict management.
6. Strengthening people's participation in the mitigation of human-wildlife conflicts.
7. Improvement of the habitat of animals in the forest including grasslands and waterholes to provide adequate food and water inside the forest itself.
8. Training, awareness creation, and research to reduce the conflict.

PHYSICAL AND FINANCIAL REQUIREMENTS: The physical requirements of each intervention such as the length and area to be covered are evaluated. The physical requirement is projected only for 5 years which will be reviewed after 5 years.

BUDGET: The total financial budget requirement for the implementation of the above proposal for 10 years is Rs.1150.00 Crore.

1. INTRODUCTION

One of the major challenges in wildlife management the world over is the negative interaction between humans and wildlife normally termed human-wildlife conflict (HWC). Conflict situations are considered inevitable in all communities where humans and wildlife share the same habitat and compete for resources. HWC is defined as any interaction between humans and wildlife that negatively impacts the social, economic, or cultural life of humans, the conservation of wildlife populations, or the environment. This definition indicates that the negative impact due to the activities of humans and wildlife could affect either.

Of late, the conflict has become graver as the intrusion of people into wildlife habitats has increased, causing changes in the land use pattern. There has also been a change of attitude as in-migrated cultivators are less tolerant towards crop raids by wildlife compared to the past few decades, mainly because of the affinity to 'developmentalism'. Developmentalism is a belief that 'development' will resolve or mitigate all the problems that emerge within the material and social processes that take place in society. It is often assumed that the ultimate responsibility for the success and failures of such development projects rests with government institutions whether outside or within the forest. This outlook is particularly prone to conflict between human life and wildlife, and the only solution is to mitigate the conflict. This requires the convergence of state and community to alter perspectives about wildlife and human existence in the forest and its fringes. Simultaneously, forest conservation should be made more effective and the binary distinction between kaadu and naadu must be reconciled. It is in this context that 'people's participation' becomes relevant and effective.

A distinction is suggested between human-wildlife conflict and agroeconomic-wildlife conflict. This distinction is pertinent mainly because crop raids are more prevalent and predominant, and the cropping pattern has a role in attracting wildlife, especially elephants and other mammals, to the human habitat. The cultivated area may also be located on the animal's habituated movement path. Conflict arises from economic losses to agriculture, including the loss of cattle through predation and the destruction of crops. The recurring visitations of wild animals to farms may give rise to somatic preoccupations that present no physical disorder but manifests as psychological conflicts among the affected populations. Psychosomatic disorders are more pronounced among the settlers rather than the tribes that have been living within the forest or at its fringes, as they may have an altogether different perspective on conflict-related matters. Therefore, the term 'human' cannot be conceived as homogenous, rather it must be viewed as heterogeneous while considering conflict situations with wildlife. This complex issue that involves eco-psycho-socio-economic factors warrants mitigation measures of different types by considering the long-term consequences of wildlife and human behaviour.

Traditionally, wildlife management decisions have relied on the biological sciences than social assessments. However, there is a need to integrate the biological and social sciences, and the most important components in such an approach are defining management in

terms of impacts on people, stakeholder involvement, and wildlife management as an adaptive process.

The term "impact" is normally considered as a special subset of the many effects resulting from interactions among people, wildlife, and wildlife habitat. A single interaction between wildlife and people could be considered as positive or negative depending on the perception of the evaluator. The same individual could have a different attitude/perception of the same interaction depending on the situation. For example, while crop depredation is a negative interaction, benefit from wildlife tourism is considered a positive interaction.

The modern approach to wildlife management accepts uncertainty and adapts to changing circumstances and new understanding developed through experience and research, in an approach called Adaptive Impact Management (AIM). This stands for the willingness to apply enough rigour to management activities to learn and improve. The affected communities normally expect immediate and undivided attention from the government machinery, desire significant involvement in management planning and decision making, and want effective solutions to their problems at no cost from the community.

The pertinent questions are

(1) What are the impacts that affect stakeholders from the damage management perspective?
(2) Is the management program focused on the impacts that matter most to stakeholders?
(3) Do the activities maximize program effectiveness by increasing positive impact and reducing negative impact?

Recognising the nature of the problem, conflict mitigation calls for a combination of adaptive management strategies based on the ecology and behaviour of the animals involved, along with site-specific spatial and temporal determinants of conflict, rather than density-based indicators such as the number of people and wildlife living together and the human socio-political and economic environments. The problem can be solved only when the local community decides that the perceived benefits derived from the presence of wildlife outweigh the perceived damage that they cause. Better transparency in decision-making and solutions that improve public tolerance of wildlife through direct benefits are thus required.

NATIONAL SCENARIO

National-level statistics indicate an annual death toll of 400 people, damage to thousands of acres of crops, and the unmeasured psychological stress on the affected community. Further, there had been retaliatory killings of several wildlife species, especially elephants, tigers, and leopards. About fifty elephants are reportedly killed to defend crops. The most 'problematic' animals as per reports were elephants, wild boar, tigers, and in some locations, leopards. Recent reports mention monkeys and peafowl as well.

Government of India records state that, during 2007-11, there have been 888 human deaths and 7381 human injury cases due to human-wildlife conflict. About 91.20 million rupees have been claimed/paid as ex-gratia to the victims. There have been 14,144 cases of livestock killing and 80,956 cases of crop damage for which an ex-gratia payment of about

137.40 million rupees was disbursed. There were 203 reports of electrocution and poisoning of animals as retaliation to human-wildlife conflicts. During the period, 31 animals were killed with official permission, considering them dangerous to human life and property.

KERALA SCENARIO

Kerala has a forest cover that approximates to 11,521.813 Sq. km, extending over 29.65% of the total geographical area of the State. With the drastic increase in human population and the resultant increase in anthropogenic pressures on forest areas, human-wildlife interaction and resultant conflict have seen a sharp rise in the recent past.

Despite the popular perception that such conflict is a new phenomenon, historical records indicate that this is wrong. Records in the early 1890s state that "Wayanad is dotted with machans" indicating that the people have resorted to tree-top guard stations to scare the animals entering the agricultural fields. An 1818 proclamation by the erstwhile ruler of Travancore specifies arms and ammunitions provided to the farmers to scare away the animals.

However, these past few years have seen escalated tensions in wildlife management in Kerala, and settlers near reserve forests and sanctuaries express a heightened sense of insecurity. The growing pressure of the ever-increasing human population, habitat loss and fragmentation, changes in cropping patterns, and increasing wildlife population are considered the major causes. An analysis of threats to biodiversity conservation and management of natural resources in various Forest Divisions of Kerala shows that human-wildlife conflict is a threat in almost all Divisions, especially in Wayanad and the rest of the northern region.

A more recent phenomenon is the proliferation of conflict issues in areas far removed from forest fringes. Due to a combination of factors like abandoned farmlands, massive accumulation of garbage, behaviour of species involved (generalists) and their breeding success, conflict creating species like wild boar, bonnet macaque, peafowl, etc., which were hitherto confined to forest fringes, have migrated to these new habitats and started successfully breeding there. They have started establishing new populations away from forest fringes. Hence serious conflict issues have moved from forest fringes to faraway settlements and even urban areas. Hence, conflict can no more be treated as a purely forest-fringe phenomenon.

ANALYSIS OF PRESENT SITUATION

Attributed Causes of Conflict

Animals, especially elephants, when confined to smaller areas by habitat loss and poaching leading to localised high densities may engage in crop-raiding in the surrounding areas. Also, farmers have abandoned communal planting and guarding activities that are used to reduce crop loss. It has been argued by the affected that the increase in wildlife population numbers (especially elephants) following improved anti-poaching measures has led to some elephants losing their fear of people. Despite inconclusive evidence, it is also held that human intrusions displace elephants, which in turn depend on crop-raiding to survive in resource-poor habitats. Other reasons attributed to conflict situations are the creation of

abundant secondary vegetation due to human activities and artificially maintained water sources that attract elephants, and the severance of traditional migration routes or movement paths by human interventions, such as canals, power installations, agriculture, settlements, and fences, leading to aggressive behaviour. Unfortunately, the various circumstances that cause mortality as a result of wildlife encounters have not been analysed as the basis of proper strategies.

There are several economic and social costs borne by communities around wildlife habitats that can outweigh the direct costs of agricultural damage and can influence the perception of the affected people. Loss of life, restrictions on movement, competition for water sources, need to guard the property (which may lead to loss of sleep), reduced school attendance (through loss of sleep, or fear of travel), poor employment opportunities, increased exposure to diseases and psychological stress, etc., are a few of effects that the affected community is subjected to for wildlife conservation. Conflict situations can cause potential harm to all involved and lead to negative human attitudes, and a decrease in human appreciation for wildlife will have detrimental effects on conservation in the long run. The goal of wildlife damage management is to increase the net benefit of wildlife for society. This may be achieved through purposeful interventions that address the adverse effects of wildlife and wildlife management that matter most to stakeholders.

Impacts

Human-wildlife conflicts manifest as human casualty and injury, crop destruction, cattle lifting, and property damage. An analysis of threats to biodiversity conservation and management of natural resources in various forest divisions of Kerala shows that human-wildlife conflict is a threat in almost all forest divisions, especially in the northern region (Fig.1 & Table 1). The degree of conflict between humans and wildlife greatly varies with the regions in the State. The figure indicates higher incidents of conflict in areas with human presence and along the fringes in certain parts.

Fig. 1 Intensity of conflict in different areas in Kerala Forests

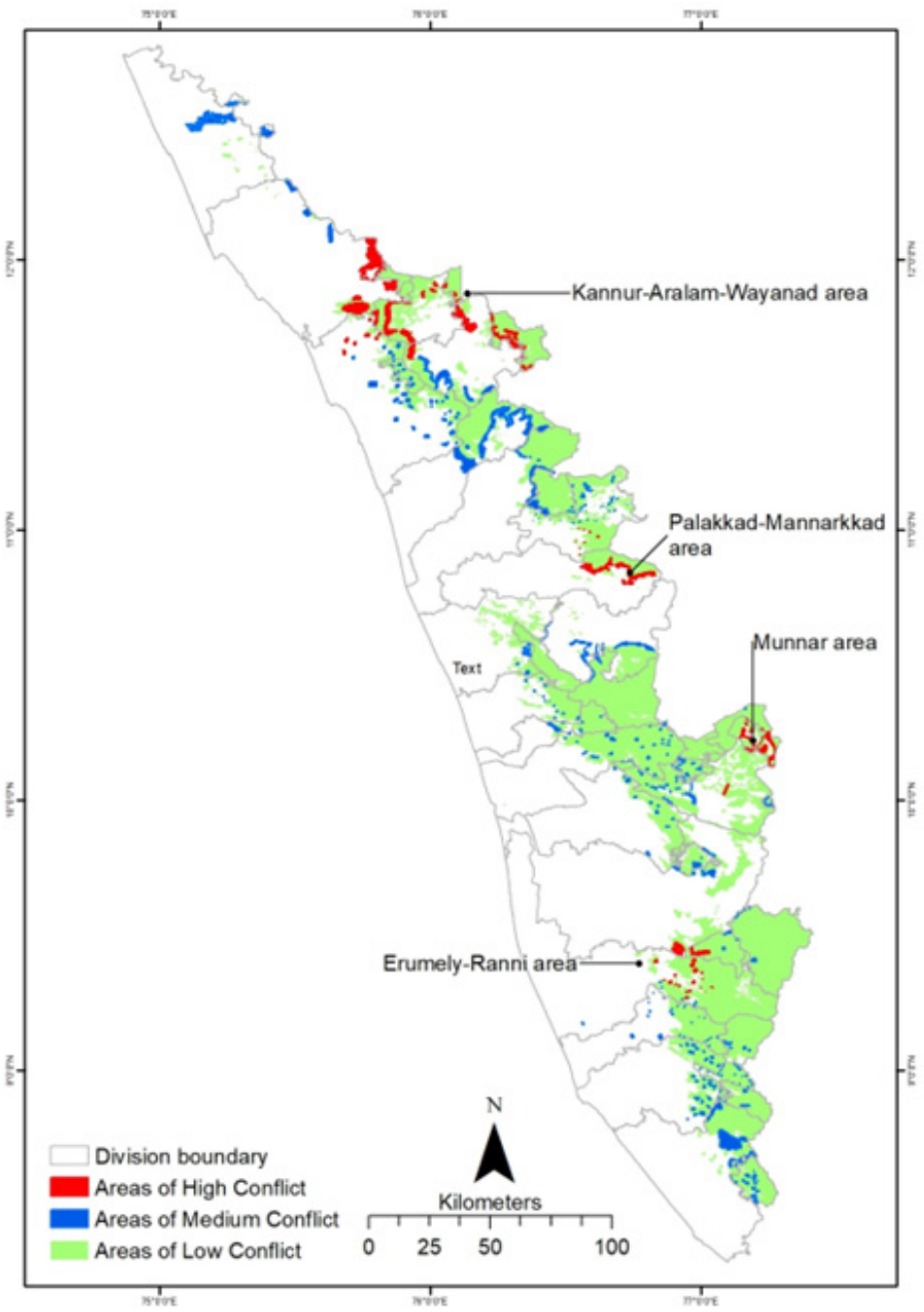


Table 1: District wise details of human deaths due to wildlife during the last 10 years

District	Number of deaths
Thiruvananthapuram	9
Kollam	6
Pathanamthitta	6
Kottayam	1
Alappuzha	0
Idukki	44
Ernakulam	8
Thrissur	87
Palakkad	59
Malappuram	18
Wayanad	44
Kozhikode	W2
Kannur	15
Kasaragod	20
Total	319

Table 2: Details of ex-gratia payment (from 2009-10 to 2020-21)

Year	No. of Applications	Ex-gratia amount Paid (In Lakhs)
2009-10	2922	119.3
2010-11	3550	201.0
2011-12	6210	454.4
2012-13	6132	460.7
2013-14	7188	659.1
2014-15	8112	782.5
2015-16	6022	681.7
2016-17	7765	963.9
2017-18	9333	1018.7
2018-19	8125	1115.0
2019-20	6859	930.06 *
2020-21	10095	842.76 *

** The figures are not final as disbursal is pending due to lack of funds*

Table 3: Details of human casualty by wildlife during 2021-22

Species	Human Casualty
Elephant	25
Snake	22
Wild boar	2
Gaur	2
Peacock	1
Total	52

The ex-gratia payments for the period 2009-10 to 2020-21, if taken as an indicator of the number of conflict-related incidents, show an increasing trend. However, this may not reflect the actual reality as claims have not been made in several cases (Table 2). Moreover, the ex-gratia payment for snake bite deaths and the increase in payment of ex-gratia for human casualties also must be taken into account. There has also been the loss of wildlife due to retaliatory killings by the affected people. This has been more evident in the case of elephants, where electrocutions and poisoning have been reported

Fig 2. Number of incidents involving elephants, wild boar, bonnet macaque, and snakes are involved (2013-14 – 2018-19

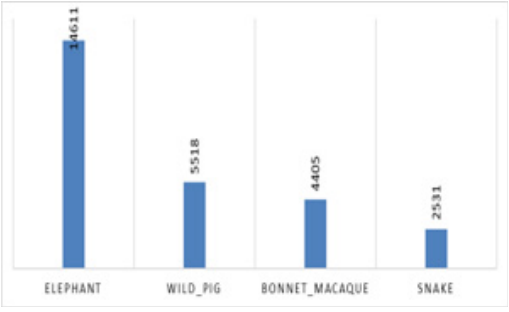


Fig 3. Number of incidents involving other animals (2013-14 – 2018-19)

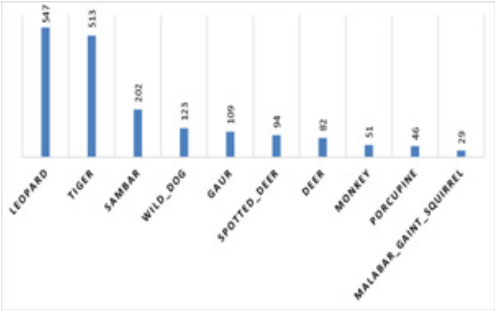


Fig. 4. Number of incidents of crops damages

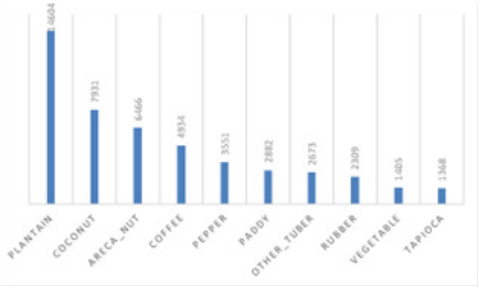
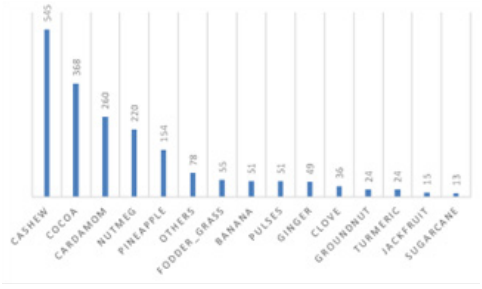


Fig. 5 Number of incidents of crops damages



Elephant, wild boar, and bonnet macaque were involved in most of the incidents (Fig. 2 and Table 3). Deaths due to snake bites are not included as it occurs mostly in human habitations. Crop damages were common in the case of coconut, areca nut, plantain, coffee, paddy, and tuber crops. The most affected crops were plantain followed by coconut and areca nut (Figs. 4 and 5). Tigers and leopards were responsible for the highest number of cattle lifting incidents and buffaloes and goats were the victims in most cases (Table 5). The amount paid in different years as ex-gratia varied as shown in Tables 6 & 7.

Table 4 Crops and the animals involved in conflict incidents (2013-14 to 2018-19)

Crops	Elephant	Bonnet macaque	Wild Pig	Sambar	Chital	Gaur
Coconut	4991	2139	886	-	-	-
Areca nut	4441	1819	608	-	-	-
Plantain	8926	2973	3074	83	50	23
Coffee	3109	1701	432	-	-	-
Paddy	1739	65	1050	-	41	-
Tubers	1103	885	802	--	-	-

Table 5 Livestock death and injury caused by different wild animals (2013-14 to 2018-19)

Species	Cattle	Buffalo	Goat	Total
Elephant	20	0	6	26
Leopard	93	2	78	173
Tiger	250	47	123	420
Snake	58	5	35	98
Wild Dog	8	2	54	64
Wild Pig	18	1	14	33
Total	447	57	310	814

Table 6. Exgratia payment for the livestock loss, 2015-16 to 2020-21

Year	No. of Applications	Amount Paid (In Lakhs)
2015-16	355	59.76
2016-17	361	76.39
2017-18	443	63.27
2018-19	379	88.39
2019-20	330	71.74
2020-21	538	72.30

Table 7. Exgratia payment for human casualty and injury

Year	No. of Application	Amount paid (in lakhs)
2015-16	513	238.98
2016-17	943	467.09
2017-18	1121	468.96
2018-19	813	501.08
2019-20	796	437.52
2020-21	2416	366.00

MITIGATION MEASURES

The most prevalent mitigation measure adopted by Kerala Forests and Wildlife Department is by erecting barriers to prevent the entry of animals into the cultivated and residential areas (Table 8). The Forest department has so far made 511.22 km of elephant-proof trenches, installed 2348.14 km of solar-powered fence, 9.7 km of solar-powered hanging fence, 66.26 km of elephant-proof walls, 10 km of rail fence, 15.12 km of kayyala, 0.20 km of bio-fence, 1.88 km of steel structures with spikes, and 32.4 km of crash guard fence.

Plans were also prepared for the voluntary relocation of settlements and a few of the identified areas have been freed of settlements through relocation (enclosures identified for relocation are given as Annexure-10). NGOs also contributed to the programme by acquiring and transferring land falling under the elephant corridor. A few attempts have also been made to capture and translocate problematic elephants and tigers using innovative methods such as deploying beehives and recorded tiger roars. The Forest Department has also formed Rapid Response Teams for immediate action in case of emergencies and working with the affected people.

The government of Kerala has also initiated a project for fencing along the forest boundary with financial assistance from Kerala Infrastructure Investment Fund Board (KIIFB), and administrative sanction was accorded for the first phase in Thiruvananthapuram, Thenmala, Punalur, Palakkad, Mannarkad, Nilambur South, Wayanad Wildlife Sanctuary, and Wayanad North. As per the project, 101.50 km of solar-powered fences, 0.41 km of elephant-proof walls, and 10 km of rail fences will be constructed in the field.

In Phase II, 136.70 km of solar-powered fences, 48 km of crash-guard steel rope fences, and 185 km of elephant-proof walls will be constructed in various divisions.

Mitigation Measures – Evaluation

Crop-raiding variables are strongly environmentally spatial rather than numerical; factors such as habitat, daytime refuges for elephants, water distribution, human settlement patterns, physical defenses, or cultural and agricultural practices may be locally important determinants of HWC levels. However, most conflict mitigations measures in the country are copied from elsewhere, mostly from Africa. An attempt is made here to evaluate the efficacies of existing mitigation measures.

Traditional methods. The traditional method involves guards on Machans (raised platforms mostly on trees) and scaring animals away using noise. This is reported to be the most effective the world over. But the cultivators find it difficult to spend the night in Machans after working all day long. However, this method is being successfully used in several places (eg. Sholayur in Attappady). While attesting to the effectiveness of the method, some of the farmers expressed difficulties in getting workers to stand guard.

Table 8: Circle-wise extent of existing preventive measures

Circle	Extent of Preventive Structures (length in kms.)								
	Elephant-proof Trenches	Elephant-proof Walls	Rail Fence	Solar Power Fence	Crash Guard Rope Fence	Hanging fence	Steel structures with spikes	Bio-fencing	Kayyala
Agasthya-vanam Biological Park	3.50	-	-	26.00	-	-	-	-	-
Central circle, Thrissur	9.41	-	-	233.41	-	-	-	-	0.28
Eastern circle, Palakkad	20.35	6.72	-	443.34	5.70	5.70	-	-	2.20
Field Director (Project Tiger), Kottayam	19.07	5.47	-	87.44	-	-	-	-	0.09
High Range Circle, Kottayam	11.44	0.22	-	153.49	1.20	-	-	-	-
Northern Circle, Kannur	215.47	31.55	-	696.18	25.50	4.00	-	-	-
Southern Circle, Kollam	14.81	1.54	-	460.74	-	-	-	-	12.55
Wildlife Circle, Palakkad	217.18	20.77	10.00	247.54	-	-	1.88	0.20	-
Grand Total	511.22	66.26	10.00	2348.14	32.40	9.70	1.88	0.20	15.12

Saree fences are used in several places in Kerala to prevent the easy entry of animals into the cultivation areas. Locally available bamboo and other materials are being used to make a thick fence making the animal entry difficult. This has been reported to be very effective against wild boar and is successfully used in Pottamaavu Kaani Colony in Arippa in Trivandrum. In certain areas, discarded fishnets have been used effectively against wild boar. But overgrowth of vegetation can destroy such fences and hence, these remain in operation for only a small period. The advantage is that the method is very cheap.

The traditional methods also include sound-based mechanisms like blasting of crackers and the replay of recorded tiger roar sound. Unfortunately, over a long period animals get used to the sound and will not be afraid of it. The lighting of fire is used to deter animals, even elephants. Of late, the reports from the farmers indicate that this method is not much effective. Dried fish hung on the border claims to have some impact on monkeys. But this was countered by others, who reported failure. Soap water is used as a deterrent in some parts of Kerala, especially against Sambar deer. Though not traditional, affected people have gone for their inventions as a solution to the conflict situations. In one place in Wayanad wildlife sanctuary, cultivation is protected by using thin steel wire fences on which used plastic water bottles are hung. The farmers claim that these would deter the animals mostly because of the sound. The Mancheeri Cholanaikan Colony in Karulai Range in Nilambur is protected on one side with trench. However, the Cholanaikans spend their nights on the rooftop which is covered with tarpaulin. This is to escape from the elephants visiting the area during nights. A crude wooden ladder is also provided to climb to the roof.

Chilly, Grease, and Pepper oil. This has been reported to be successful in Africa and attempted in several places in the country. Smearing rope with chilly and grease and tying it along the border did not work in high-rainfall areas. The smell of chilli alone may not be fully effective, and some simple supplementary efforts are still needed. It has been suggested that a 5 m wide area clear of any thick vegetation should surround crop fields to help with elephant detection. A sounding device hung on the chilli fence can alert sleeping farmers when an elephant touches the string barrier. Farmers should maintain good vigilance during crop maturity (eg. taking turns to man watchtowers).

Dung and chilly smoke. Hand-held pepper sprays (aerosols) or smoke canister bombs fired from simple mortar-like launchers and pepper dung (chilli-dung) burnt to produce a noxious smoke have also been tried in other parts of Asia and was found to be partially effective as the success depends on the direction of the wind.

Beehives as a deterrent. Following the success stories in Kenya, the concept of beehive fences has attracted enthusiastic conservationists in some parts of Asia. This was also because of the income expected from the honey and the affordability to small-scale farmers. However, this method failed in Indonesia and Sri Lanka. The experiments in Thailand showed some positive responses in the beginning but failed subsequently because the elephants got used to the buzz sound of the bees. In effect, the very purpose of conflict mitigation failed.

There was also an attempt in Wayanad under the leadership of ATMA, but the elephants dismantled it. There was a successful attempt along the borders of the Peechi-Vazhani

Wildlife Sanctuary. However, in this case, the beehive fence was supported by a solar power fence. So, the effect of the solar fencing must be considered as well. An experiment done by a researcher in KFRI reported success but external factors were not considered. One must be cautious while implementing beehive fence in areas where sloth bears are present. Honey may attract bears, thereby adding to the problem.

Bio-fencing. Agave, reportedly avoided by elephants, and xerophytes might help. Unfortunately, the thornless agave is not effective and is eaten by elephants. The thorny one is reported to have some effect. There had been suggestions a few years back to try Salaca, an exotic species resembling an oil palm. However, studies mention that it is used as elephant food in countries such as Myanmar; even the fruits of the palm, when ripe will attract other animals. Unfortunately, there are no reports of success from any part of India. However, there have been some reports of the effectiveness of some plants grown along with vegetative fences.

Elephant-proof Trenches. Elephant-proof Trenches (EPTs) are an old practice in several parts of the country. One of the reasons for acceptance of the trench as a mitigation method was the possibility of employment generation, but they have hence been replaced by machines. EPTs are not effective in heavy rainfall areas and depend a lot on soil conditions. In addition, the method needs heavy initial investment and if not maintained annually, they become ineffective. Elephants learn to kick in the sides of trenches and cross them and are also undeterred by the narrow stretches of water. An unpublished report based on observations by a Soil Conservation Officer in Wayanad indicates that the trenches disturb the first-order streams in the area and rainwater is artificially diverted leading to scarcity in the marshy areas of Wayanad. So, there is also an unaccounted environmental cost involved in the method. Stone-pitched trenches, which is the latest trend, are expensive (95 lakhs/ km), q and quarrying for the construction material results in an environmental cost. These trenches are a permanent scar on the earth, wherever it is done. One of the major issues with ditches or trenches is the high investment cost; both for construction and maintenance, the latter because of vulnerability to soil erosion.

The trench near Kuruva (at Chegady) was probably the first in Wayanad area. Appappara colony in North Wayanad Forest Division has a trench around it, but it is damaged and is reinforced with a solar fence, which was also not so effective due to poor maintenance. Wayanad Wildlife Division has trenches in almost all the ranges and is mostly reinforced either with a wall or solar fence. Trenches are ineffective in most of Wayanad because of the marshy land area (kollies). In Nulpuzha Panchayath in Wayanad, the trenches dug out in 2000 are reported to have failed because of the loose soil. The 35.5 km long trench in the area was crossed by elephants. The marshy areas were left out due to obvious reasons. All along the trench, a solar power fence was installed later. It also became non-functional due to the lack of maintenance, and almost 80% is ineffective.

There was a trench in the Thoova area from Marappalam to Kodankarappallam in Agali in Mannarkad Division. Most of the trenches dug as a part of mitigation measures in different parts of the State did not meet the objective either because they were incomplete or because

maintenance was lacking. In some places, the failed trenches had solar-powered fences or walls to augment the effectiveness of the trench.

Solar Power Fence. Solar power fencing is a technologically effective elephant deterrent. This is one of the most popular mitigation measures. Fences are successful only if it encircles the layout to avoid the entry of wildlife, and where scrupulous daily maintenance is possible. Clearing the vegetation near the fence regularly and using strong supporting poles can help. A team of people patrolling along the fence at night (human presence) will add to the effectiveness. This has been proved in a few places in Wayanad and at least one place in Punalur Division.

Wayanad is the area with the highest number of solar power fenced locations. Solar-powered fences with hanging wires have become popular and are reported to be more effective because it is beyond the reach of elephants thereby avoiding frequent damage.

Several farmers have installed solar power fences in the Vadakkanad area though the larger area is protected with solar fences erected by the department. All privately installed fences are effective. However, there are occasional damages by elephants. These are immediately repaired by the owners. A very effective solar power fence was seen in the Sholayur area of Agali Range of the Mannarkad Division, installed at a cheaper rate by a private individual to protect his farm. There are instances of installing power fences by private individuals in a haphazard manner, preventing the movement of elephants and creating problems in adjacent areas. Such examples could be seen in Munnar also.

The maintenance deficiencies are almost always caused by weak institutional arrangements (related to contacts with contracts, wages, work schedules, disputes, etc.) rather than by any technological shortcomings. Non-maintenance mostly contributes to the failure of solar power fences. However, there are success stories too. The fences installed in 2006 around the Mampazhathara tribal settlement in Punalur Forest Division has been functional for many years due to regular maintenance by the beneficiaries.

Crash Guard Rope Fence: This has been attempted first in Mankulam and was found effective till the 2018 flood when some of the pillars were damaged. These are being repaired and the beneficiaries have a very high opinion about the effectiveness. It is considered a good solution, especially in marshes. However, it must be installed in straight lines, and the zigzag borders delineating forest and private land are not well suited for this unless adjustments are made to straighten the line. Since the fence is normally taken as the boundary, there could be disputes in the future. The cost varies according to the soil types and could range from Rs.49 to Rs 58 lakhs per kilometer.

Kayyala and Elephant-proof Walls. This is one of the latest trends in Kerala. The implementation cost is very high (140 lakhs/km) and the building materials are procured at an environmental cost. According to one report, about 9000 cubic feet of granite stone are required for the construction of the 100 m wall.

This was implemented in Aralam, where the height had to be ultimately increased and was still not successful as the work could not be at the required length (only 5.065 km),

obviously due to lack of funds and high cost. A 10 km wall was erected on the southern side of Aralam (from Valayamchal to Kariamkappu) and is reported to be successful in preventing elephants, but as a consequence, the elephant problem has shifted to other areas.

There are other areas in the entire Wayanad, where walls have been constructed mostly to block the entry of animals through marshy areas. There are also proposals for new ones. However, there are reports of elephants crossing even 2 m high walls in Wayanad. At least in one place in Nilambur South Division, the wall was reinforced with a strand of solar power fence on the top.

Steel structures with spikes. Spikes fixed on concrete structures have been tried in Aralam. This was part of a 10.5 km barrier along the boundary with the human habitation. This boundary also has 5.065 km stone walls and a 3.55 km trench. A solar power fence was also seen parallel to the steel fence with spikes. But the steel fence was broken open by a herd of elephants crossing to the nearby areas. This is one of the worst conflict mitigation measures as this would lead to injury to the elephants while attempting to break open. Moreover, it has been proven to be ineffective. In the South Wayanad area, very strong steel gates with spikes fixed have been complemented by solar power lines. However, the strong steel gates were bent by elephants.

Radio-collaring elephants and early warning. A section of the scientific community claims that problematic animals could be captured, and radio-collared for monitoring their movement. This requires identifying and collaring the individual problem animal, receiving GPS locations via satellite or GSM technology, mapping the location on GIS maps, and informing the farmers who are at risk. The cost is very high. This would generate information on temporal and spatial patterns of crop-raiding, possible contributory factors, the effectiveness of preventive methods, etc. Moreover, if elephants are not residents, but rather seasonal visitors, theoretically it is possible to monitor the movements of individual crop raiders through GPS satellite radio collars and warn villagers of their presence. It is not applicable in areas with a significant number of animals. These measures did not help in Wayanad and North Bengal.

Use of drones for scaring. A recent study finds that small unmanned aerial vehicles, also known as UAVs or drones, help reduce human-wildlife conflict. Since late 2014, quadcopter drones have been used by rangers in Tanzania to safely shepherd elephants away from farms and communities. Elephants seem to keep away because of the buzzing sound of the drones. There have been some pilot studies in Sri Lanka last year. Though found to be effective during field trials, according to the researchers in Sri Lanka, there is every possibility of elephants getting used to drones.

In India, the Air Force had reportedly used an aircraft to scare away elephants from its Tezpur station in Assam some years back but this was not found to be effective. The West Bengal Forest Department used a drone in July 2021 in the Bankura district to locate and monitor a tusker proclaimed as rogue. It was reportedly not of much use as compared to the trained elephants (kumkis) and on-foot parties. There are also regulations from the Director General of Civil Aviation which have banned civilian use of drones in some areas in the

country till finalisation of guidelines. In Kerala, drones were used to locate the strayed-out herd near Aralam.

Early warning electronic systems. Early warning systems have been functional in Wayanad Wildlife Sanctuary. Elephant Information Groups were formed in different areas and information on elephant sightings are received by the Divisional Control Room managed by RRT. These sightings are forwarded to the concerned area staff and watchers for verification and displayed in the concerned area. The display boards are installed at four places, one in each range. Wherever there is no display board, the information is passed through SMS alerts. Unfortunately, the display boards were removed because of the complaints from the public. The SMS alert system is currently not functional and is waiting for updating of the software.

The South Wayanad Forest Division has formed a WhatsApp Group and information on elephant sightings is passed on to the members so that people could be alerted on the presence of elephants, if any in the area. The Divisional Forest Officer and some of the stakeholders were very happy with the arrangement.

Translocation of Problem Animal. Capture and translocation of problem animal had been suggested on many occasions. Experience in Hassan in Karnataka and once in Kulathupuzha suggest that the translocation does not help. The released animal returns to the site of capture or recreates the same problem at their release site. There is also the possibility that a proportion of the translocated animals die either due to the stress of capture, endangers a resident population through the introduction of disease, or increase competition for territory or food. Removal of the animals from the 'problem area' does not reduce the number of raiders, because other recruits in the 'occasional raider' subpopulation merely replace them. The claimed success story of the capture of "Vadakkanad Komban" (Vadakkanad tusker) is reported to have been successful. But there is no reduction in crop-raiding in the area even after the capture. There were differing views on the capture of this animal even among the general public in Wayanad.

In the case of tigers and leopards, the capture of problem animals in cages used to be the usual practice in the State. Several leopards have been trapped in different places and were normally released in forests. However, the practice is currently discontinued, and Government of India Guidelines are followed as far as possible. One of the major issues currently faced by the Forest Department is regarding cattle-lifting by tigers, especially in Wayanad. Most of these were captured and shifted to zoos or the animal died due to various reasons.

Monkey, especially the bonnet is considered to be a problem even in villages far away from forested areas. The Department attempted to solve the "monkey menace" in Sasthamkotta near Quilon by trapping the so-called "market monkeys" and releasing them in the forest areas in Kulathupuzha. The vacant space in Sasthamkotta is now occupied by the "temple monkeys" from the same area. The people of Kulathupuzha, where the trapped ones were released complain of the monkey problem in their area. The problem was shifted from one area to the other and the problem in the source remains. There had also been some attempts

to capture bonnets from Kalpetta of Wayanad. The process resulted in mortality and was discontinued due to public protest.

Unpalatable crops as a buffer. Growing unpalatable crops as a buffer to farmlands is mainly aimed to provide alternate revenue to farmers in high human-elephant conflict zones (coffee, lemon, chilly, etc.) The use of these crops as buffers is restricted by the type of soil, rainfall, topography, availability of water, and market prices. Change of cropping pattern from traditional livelihood farming to alternate crops by farmers does not seem to be practical and is unacceptable to cultivators in many areas. Pilot studies on the effects of alternate crops such as capsicum and lemon, with the introduction of Citrus spp as elephant deterrents along with other economically viable crops, are in experimental stages in the Chirang-Ripu Elephant Reserve in Manas National Park. Medicinal and Aromatic Plants (MAPs) are being tried in Nepal. Chilly was planted in the buffer zone in Hassan in Karnataka. Elephants damaged the plants by trampling.

Providing Water and Increasing Food Availability in the Forest. Most often, the straying of animals into the habitation is attributed to the lack of food and water in the forest. The farmers in Wayanad hold the view that this is the major cause and according to them, it is due to the plantations of eucalyptus and teak in the area. This need not be true though some of the plantation areas do lack undergrowth. But the records from different parts of the State show that most of the crop-raiding is during the rainy season. It is the crops during the season, which attract animals, especially elephants. A study in Wayanad also indicates good availability of food inside the forest plantations. For example, the teak plantation in most of the Kurichiat range is as good as natural forests. However, there are issues in certain areas in some of the ranges. The lack of food in these areas could be attributed to cattle grazing and, in some places, to fire degrading the forests. According to field level reports from farmers and people's representatives, about 80% of the estimated 10,000 cattle graze in the forests. Interestingly, these are given to the tribal people by a meat processing unit on a buy-back system and the Forest Department is helpless since tribals are involved. A more recent phenomenon is the rapid proliferation of a weed, Senna.

An analysis of water availability in Wayanad has indicated that there is at least one water hole per 2.5 sq. km of the forest areas. The water holes are currently monitored for information on their use by wildlife. The non-availability of water in the area does not hold for most of the locations.

Voluntary Relocation. In Wayanad District, people were relocated from Arakunji, Golur, Ammavayal, Eswarakolli, Vellakod, and Kottankara settlements and most of the people from certain other settlements such as Chetty-Alathur, Kurchitat, Narimundakolli, Pambumkolli, Pankalan, Kolot, Pilakavu, and Puthur-Manimuda have also been relocated. Out of 646 eligible families in 14 settlements in Wayanad Wildlife Sanctuary, 382 families have been relocated so far. The relocation process is going on. Under the Rebuild Kerala Development projects, as part of relocating non-tribal settlements, 677 families have been identified in 22 settlements coming under 4 forest divisions (Annexure 9). In the first phase, 50 families in Kattilappara settlement coming under Shendurney Wildlife Sanctuary have been paid the first installment of money.

Elephant Corridors. The Wildlife Trust of India has identified nine elephant corridors in Kerala in the book "Right of Passage- Elephant Corridors of India". The locations of the identified corridors as per this publication are as given in Annexure 10.

In the case of the Nilambur Kovilakam-New Amarambalam elephant corridor and Mudumalai – Nilambur via O' Valley (Ouchterlony Valley) corridor, no acquisition is needed. For establishing the Thirunelli-Kudrakode elephant corridor, 9.63 acres of the acquired area have been declared as a Wildlife sanctuary as an extension to the existing Wayanad Wildlife Sanctuary & the remaining 8.37 acres as Reserve Forest. In the case of the Kottiyoor–Periya elephant corridor, the acquisition is partly done and in the case of the remaining corridors, the process of acquiring land is in progress.

Wildlife Population Regulation. One of the major contentions of the public on the conflict issue is mostly related to the increase in the wildlife population. Based on the local knowledge and experience, they claim that there is an increase in the population of elephants, wild boar, and tigers (Wayanad). Unfortunately, there had been no attempts to analyse the population estimation data for the different landscapes to prove/disprove the claim.

The management interventions in the State and even in the country have been limited to habitat management under the assumption that addressing the habitat degradation issue would help in the recovery of habitat and persuade the animals to stay back in their natural habitat. Population management has not been on the agenda mostly because of ethical issues and data deficiencies.

Immuno-contraceptives are reported to be successfully used to control the elephant population in Africa. Though there are demands from wildlife managers to use the technique in India, there are issues of standardization of the method in Asian Elephants. Surgical sterilisation is reported to be successful in monkeys. The results of the attempts in Wayanad are yet to be known. For the method to be successful, at least 70% of the females need to be sterilised and there should be continuous monitoring of the population. Moreover, the population should be a closed one with no possibility of immigration from nearby troops.

Jana Jagratha Samithis. Jana Jagratha Samithis are formed with people's representatives, farmers' representatives, and Forest Range Officers, and function at the Panchayath level. There have been initiatives to form ward-level Samithis where local issues are discussed. These Samithis are effective because conflict issues are discussed in detail and solutions suggested. Moreover, public support is ensured in meeting the challenge. There are suggestions to form a district-level statutory committee with the District Panchayath President, Collector, and representatives of Panchayath level Jana Jagratha Samithis (Details of Jana Jagratha Samithis are given in Annexure 6).

There are several LSGs, who have contributed to the implementation of conflict mitigation measures. There could be a government order in this respect empowering the Panchayaths for financial support.

There had been initiatives from the Forest Department to organise interactive meetings

with People's Representatives and media persons. Such meetings were regularly organised in Wayanad to exchange ideas. Moreover, a lot of misconceptions about wildlife management issues get cleared. Such interactions may be organized in high conflict areas.

Rapid Response Team. Presence of a team of people led by well-trained officials and equipped to deal with conflict situations often pacify agitated people and help assure them safety. This is equivalent to guarding by making the human presence felt. Unfortunately, the attempt is after the occurrence of incidence unlike guarding, where precautionary measures are taken. About thirteen RRTs are in place in different locations in the State (Details are given in Annexure 5). The members of the team are handpicked and trained on various aspects of wildlife management including the biology and behaviour of animals involved in the conflict, public relations, and also on handling different situations. However, they are not well equipped and are prone to fatigue from having to work constantly.

Payment of Ex-gratia. This is an approach popular with communities adversely affected by wild animals. Unfortunately, the word "compensation" replaced has ex-gratia and people are given the wrong message that the entire damage will be compensated. Ex-gratia also reduces the incentive for self-defense by farmers and therefore, could even exacerbate the scale of the problem. Moreover, ex-gratia is unable to decrease the level of the problem and the cause of the problem when the cause of the problem remains unaddressed. There had also been an argument that ex-gratia does not address the unquantifiable social 'opportunity costs' borne by the affected people. Clinical ethnographic inquiries conducted in Assam make visible the hidden mental health dimensions of the human-elephant conflict. According to the case study, the health impact of the conflicts penetrates far deeper than the immediate physical threat from elephants, worsens pre-existing mental illness of marginalized people, and leads to newer psychiatric and social pathologies. Timely payment is also crucial to get conservation support from the people.

The policy of paying ex-gratia to snakebite victims even outside the forest areas puts severe economic pressure on the department by spending a major share of the meagre financial resources for the purpose. One of the major complaints from the public is the cumbersome procedures involved in claiming the ex-gratia and the delay in payments. Some farmers feel that the procedure can be simplified and entrusted to the concerned Grama Panchayaths. There are a few others who do not consider this a good suggestion (Rates of ex-gratia paid are given in Annexures 7 and 8).

Policies and Provisions in Act. The Wildlife Action Plan, 2002 mentions the mitigation of human-wildlife conflict in and around PAs, use of professional and innovative approaches, provision of adequate resources for ex-gratia payment and other means of managing the problem, research, and good practices as well as policies, programs, and schemes. Wildlife Protection Act, 1972 provides the necessary guidance to deal with the problem of human-wildlife conflict under sections 11 (1) (a) and (b). These sections empower Chief Wildlife Warden to deal with the problem animal under schedules I to III and IV. Under section 62 of the Act, the Central Government is empowered to notify wild animals other than those specified in Schedule I or Part 2 of Schedule II to be vermin for a specific period

and area. The government of India has also issued a series of advisories and guidelines for human-wildlife conflict management. Recently, the government of India has also notified some problematic animals (nilgai, wild boar, and rhesus macaque) as vermin per section 62 of WLPA as per the request received from the states of Bihar, Uttarakhand, and Himachal Pradesh. Kerala Forests and Wildlife Department has also formed a committee to look into the problem of wild boar infestations and suggested some measures.

The Government policy of relocating people to areas adjacent to wildlife habitats has contributed to /aggravated the conflict situations. The best examples are Aralam and Anayirangal in Munnar. Lands were allotted to people from the Aralam farm adjacent to Aralam Wildlife Sanctuary leading to the escalation of conflict incidents. A similar action by the Government in Anayirangal brought people to the elephant habitat depriving the animals of the available areas. People now complain of damage to life and property due to wildlife. The original inhabitants of the area were in five settlements and did not have many complaints. These Muduvan communities were in Chembakathozhukudy, Kozhippannakudy, Tankukudy, Pachappulkudy, and Aduvilanthankudi. In 2003, the land was assigned to people from different areas and they settled in 301 Acre Colony, 80 Acre Colony, Panthadikkalam Colony, Vilakku Colony, and Suryanelli Colony. Most of the settlers were not familiar with wildlife areas and have a comparatively low tolerance for animal presence.

In addition to the arrival of new settlers, Anayirangal and adjacent areas experienced large-scale developmental activities related to tourism. Several tourism facilities such as resorts came up and a network of roads developed for access to these facilities. The presence of estates, roads, resorts, and the constant movement of people resulted in the fragmentation of an already shrunken area. All these activities aggravated the issues leading to changes in animal behaviour and aggression leading even to human loss. It will be interesting to study the ecological and socio-economic changes in these two areas, Aralam and Anayirangal to understand the pre-and post-settler scenario.

RECOMMENDATIONS

It is important to understand that human-wildlife conflicts are largely a human-induced phenomenon, compounded by the species-specific behavioural ecology of animals and external environmental factors. Thus, mitigation measures for all negative interactions should be developed on the basic premise of meaningful engagement with and participation of all primary stakeholders, especially local communities. (India's National Wildlife Action Plan - 2017-2031).

Complete elimination of human-wildlife conflict is not possible. But well planned, integrated approaches to managing it can reduce negative interactions and lead to a form of co-existence between people and animals. Kerala's wild elephants embody the problem best; which are restricted to fragmented and depleted habitats.

Species-specific and zone-specific solutions should be the principle behind the approach to mitigate negative interactions between humans and wildlife in Kerala.

An attempt is made here to differentiate the problem animals and problem zones as given below:

Problem Animals

Wild animals that thrive in forest habitat

Elephant, tiger, leopard, wild dog, sambar, sloth bear, gaur, and spotted deer are seen mostly in the forested areas. Kerala Forest Department is primarily responsible for mitigating the negative interactions caused by the above animals.

Wild animals that thrive in human habitations

Wild boar, bonnet macaque, snake, peacock, porcupine, and Malabar giant squirrel have been reported in human habitations. Mitigation becomes the collective responsibility of the Forest Department, VSS/EDCs, Oorukoottam, Local Self Government Institutions, Agricultural Departments, and other Departments related to infrastructure development of the local communities.

Problem Zones

Forest fringe zone (up to 2 km from forest boundary)

The approach to mitigate human-animal conflict shall be based on the needs of the people living in the fringe areas. Forest Department along with VSS/EDCs will collaborate with Panchayaths in managing conflict-related issues in the zone.

Far forest zone/Exclusion zone (Beyond 2 km from forest boundary)

The approach to mitigate human-animal conflict in these areas shall be entirely different from the earlier approach of invoking provisions contained in Section 11(1)(b) of the Wildlife (Protection) Act, 1972 in the case of wild pigs. The primary responsibility of implementing works and activities for managing conflicts is vested in Grama Panchayaths.

As the Forest Department has already moved into integrated landscape-based management, it becomes imperative that conflicts are also reported at the level of landscapes. Disappearance/blockage of a particular corridor may manifest as conflict elsewhere (Impact of linear infrastructure like roads, canals, penstocks, etc. can be visualized only at landscape level). Hence, it is suggested that there should be a conflict mitigation strategy at the level of landscapes identifying conflict zone, corridors, habitat quality, water availability, community profile, etc. This should be implemented through division-level conflict plans; which were prepared based on the zoning of conflict areas through its intensity, prioritization for intervention, collaboration with Panchayaths and capacity strengthening, appropriate research, identification of volunteers and resource persons, deployment of rapid response teams, etc. Micro-level plans are to be prepared for each conflict zone that would include identification and rationalization of existing barriers, selection and training of PRTs and RRTs, implementation of early warning systems, real-time reporting of a conflict situation, and post-conflict mopping up (eg., If we take Anayirangal, the issue can be resolved only through a combination of policy decisions including social, scientific, and ecological studies, rationalization of barriers, awareness creation among stakeholders, the intervention of PRTs and RRTs, early warning system, etc.).

The Forest Department had invited opinions and suggestions from the Members of Legislative Assembly, other people's representatives, scientists, NGOs, forest officials, and the public through the publication of notices inviting suggestions on the website, widely circulated newspapers, and through All India Radio. A total of about 1600 suggestions were received from various sections of society. The suggestions and opinions so received from various stakeholders were analyzed and legally permissible and innovative ones were incorporated into the report.

The suggestions from these could be summarised as follows. The priority should be to prevent wild animals from entering agricultural fields and human habitations. To accomplish this, various deterrent measures including physical barriers should be created. To supplement the deterrent measures, early warning systems shall be required to enable the community to be warned early for driving away wild animals before any casualties and damages. Despite this, if wild animals enter agricultural fields and human habitations, repellent measures should be used as a last resort to drive away the animals. If the measures are not successful and wild animals damage crops and property, enabling mechanism is required to pay the ex-gratia to the affected persons immediately.

1. Present report recommends the following broad strategies to mitigate the human-wildlife conflict in the State.
2. Creation of physical barriers to prevent animals from entering the human habitations and crop fields. The barriers will be used judiciously taking the landscape approach so as not to disrupt the natural movement of the animals, which is essential for the conservation of elephants in the landscape
3. Adoption of measures to avoid the interface between humans and wild animals.
4. Developing infrastructure and human resources for effective mitigation of human-wildlife conflict.
5. Providing crop insurance, life insurance, and ex-gratia to the victims of wildlife damage.
6. Involving Panchayaths and line departments to assist in conflict management.
7. Strengthening people's participation in mitigation of human-wildlife conflict.
8. Improvement of the habitat of animals in the forest including grasslands and waterholes to provide adequate food and water inside the forest itself.
9. Training, awareness creation, and research to reduce the conflict.

SPECIFIC RECOMMENDATIONS TO DEAL WITH PROBLEMATIC ANIMALS

Dealing with elephant problem

The human-elephant interaction is the result of the pattern of use of available habitats by people and elephants. Protected areas in India are estimated to cover only 22% of elephant habitat. Only a few populations of elephants in India live in continuous habitats, while others live in highly fragmented forests interspersed with a very high human population. Studies reveal that crop-raiding by male elephants was significantly higher in comparison to female herds. Habitual crop-raiding males are reported from several countries. Usually, elephants raid crops because they are more palatable, more nutritious, and have lower secondary defenses than wild browse plants, while bulls in particular may take advantage

of the easy availability of crops and stored grains. Adult and sub-adult male elephants are responsible for most of the human attacks and deaths. Whether elephants are forced by circumstances or make the best of the opportunity, the interaction between elephants and humans sometimes leads to accidental injuries and deaths of both the elephants and humans.

From 2011 to 2021 elephants have been responsible for 190 human deaths in Kerala. Elephants are also responsible for the deaths of cattle. Ex-gratia is being given for crop damage as well as human/cattle death/injuries caused by elephants. Several preventive measures were taken to mitigate the human-elephant conflict in the State. Details of existing measures to reduce human-wildlife conflict are given in Annexures – 2 and 3.

Elephant-proof Trench (EPT). This method may be treated as a traditional method since this has been an old practice in several parts of the country. One of the reasons for the acceptance of the trench as a mitigation measure was because of its effectiveness against larger wild animals like an elephant. While implementing this method, the parameters suggested in the guidelines circulated by the Wildlife Institute of India should be strictly followed.

Solar Power Fence (SPF). SPF is a very effective elephant deterrent if properly installed and maintained. SPF acts as a psychological barrier but does not provide a long-term solution, especially in areas with habitual crop-raiding elephants. The initial planning, layout, and design of the fence are very important for preventing the target species. While implementing this method, the parameters suggested in the guidelines circulated by the Wildlife Institute of India should be strictly followed.

Hanging Solar Power Fence (HSPF). This is a modified version of SPF, protected using wires hanging from the mainline, which is kept beyond the reach of elephants. HSPF consists of 15- to 20-feet high pillars every 15 feet from where stainless-steel wires hang. The wires that dangle from these pillars do not touch the ground. Solar power is passed through the wires as in the case of standard SPF and when an elephant touches the wires, it gets mild shocks that do not pose any danger to its life. Such fences are laid across the forest fringes to prevent the entry of elephants from the woods. This is claimed to be very effective as damages are almost nil and very popular in states such as Karnataka and Tamil Nadu. This method has been suggested as a good solution, especially in marshy areas and high conflict zone areas. Hence, HSPF is proposed in all the Forest Circles to supplement the other physical barriers. HSPF is less expensive to build but is effective only if maintained properly.

Bio-fence. The plant Agave, reportedly avoided by elephants, and xerophytes, lemon, chillies, date palm, etc. have been tried in various parts of the world to prevent animals. Unfortunately, the thornless agave is not effective and is eaten by elephants. Closely planted Caesalpinia sappan has been reported to be effective against wild pigs and other small animals. Similarly, some varieties of strong green chillies are planted to keep the animals away. The planting of Palmyrah palm need to be tested in the field to see its efficacy. It is proposed to take up bio-fencing works in selected areas, and the field team can consult with

local communities and decide the type of species suitable to that locality.

Early-warning Electronic Systems. Early warning systems are presently functional mainly in the Munnar region. Elephant Information Groups are formed in different areas and information on elephant sightings is communicated to the control room managed by RRT. These sightings are forwarded to the concerned area staff and watchers for verification of facts and if found correct, are passed through SMS alerts to all the registered users. People moving in the area can make use of this information to avoid the places where the elephants are moving. The message alerts are given in the form of text in local languages as well as a voice message to help those who cannot read. This is found to be very effective in high elephant conflict areas and it is recommended to be proposed to set up more such units in the State.

Use of drones for scaring/monitoring. A recent study finds that small unmanned aerial vehicles, also known as UAVs or drones help in reducing HWC, especially in monitoring and reporting. Elephants seem to keep away because of the buzzing sound the drones make. There have been some pilot studies in Sri Lanka that reported that they were found to be effective during field trials. The West Bengal Forest Department used a drone in Bankura district to locate and monitor a tusker proclaimed as a rogue. In Kerala, a drone was once used in limited locations to locate stray tigers and elephants. Drones can be used in many other regular management activities like fire management, wildlife population estimation, etc. Hence, the procurement of drones has been proposed for all the forest divisions for enhanced surveillance.

Radio-collaring of problematic animals. The problematic animals and habitual crop raiders may be captured with the permission of the Chief Wildlife Warden and radio-collared for monitoring their movement. This requires identifying and collaring individual problem animals, receiving GPS locations via satellite or GSM technology, mapping the location on GIS maps, and informing the people who are at risk. This would generate information on temporal and spatial patterns of crop-raiding, possible contributory factors, the effectiveness of preventive methods, etc. Moreover, if elephants are not residents, only seasonal visitors, theoretically it is possible to monitor the movements of individual crop raiders through GPS satellite radio collars and warn villagers of their presence. Hence, it is proposed to procure eight radio-telemetries with satellite subscriptions and hand-held receivers. The data thus collected using the satellite will appropriately be collated and analysed for devising strategies to mitigate human-wildlife conflict in the State. The procured radio collars may be provided to the Tiger Conservation Foundations (TCFs) existing in the state and utilized as and when required. The necessary financial requirement for this component is to be included in the budget provision.

Translocation of problematic animals. Capture and translocation of problem animals had been suggested on many occasions for species that create havoc in the human habitations. Though experience elsewhere suggests that the translocation does not always help as the released ones return to the site of capture or recreate the same problem at their release site, this method is suggested as a last resort solution only for animals that create serious

problems in the human habitations. Translocation of elephants especially in the Wayanad region proved to reduce the problem as the visits of the identified habitual crop raiders to the affected area have come down considerably. But the possibility of new entrants into the space vacated cannot be ruled out. In the case of tigers and leopards, the capture of problem animals in cages is a regular practice in the State. Monkeys, especially the Bonnet macaque is considered to be a problem even in villages far away from forested areas. Hence, this method was suggested in many Forest Divisions in the state. For the capture and translocation of animals, especially the endangered species, various guidelines issued by both State and Central Governments should be strictly adopted. Necessary budgetary provision is required to deal with the translocation of problem animals.

Dealing with wild pig menace. Kerala currently faces major threats from the growing wild pig (Sus scrofa) population and their number and range are found to have increased outside forest areas with consequent widespread negative impacts. Wild pigs are robust animals with high reproductive rates and destructive behaviours. The ecological and economic impacts of this species include the spread of diseases, and damage to crops, amenities, and infrastructures. It is a fact that the populations of conflicting animals, especially wild pigs that are prolific breeders, rapidly increase in the surrounding region of human habitations. The government has already issued an enabling order for the culling of wild pigs that cause problems in human-dominated landscapes in the State. To effectively remove the wild pigs from the human habitations following measures are proposed:

1. The wild pig populations permanently housed in areas away from the forest should be captured using drop-door cages. The pigs so captured should be either culled or released in tiger-bearing forests to augment the prey base.
2. In the farmlands of forest fringes, the following strategies are recommended:

a. Crops such as mango ginger (Curcuma amada Roxb.), which wild pigs detest, have been profitably planted in fields bordering forests. The fresh mango ginger sells at a current price of around Rs.90/- per kg in the market. Another crop that can be tried in suitable areas is lemon. It is financially attractive and not raided by any animals.

b. Wild pig-proof fences are another option to protect the farmlands on the forest fringes. A chain-link fence with 15 cm masonry at the bottom of the fence is the most effective to stop animals like wild pigs. The cost will be approximate Rs.550/- running meter for a 1.5-meter-high chain link mesh fence with 12-gauge tempered wire.

c. Pigs are large, strong, and adept at breaching many standard fencing designs. Fences designed specifically to completely block access by wild boar must be considerably stronger than normal fencing and may have different design features.

d. Electrical (solar-powered) wire fencing along with polythene sheets is another successful barrier against wild pigs.

e. Trained guard dogs are effective in chasing the pigs away.

f. In the rubber plantations of central Kerala, white-colored plastic sheet fences have proved to create panic in the herd.

g. Compatible crop combinations are fundamental for the success of any farming activity. Likewise, cropping patterns should also consider the likely threat perceptions from

animals including wild pigs in an area.

h. In certain areas, discarded fishnets have been used effectively against wild boar. But overgrowing vegetation will disrupt it and hence will be useful for a much smaller period. The major advantage of this method is that it is very cheap.

Dealing with monkey menace

The Bonnet macaque (Macaca radiata) creates havoc in the human settlements outside forest areas as well in the towns. Though no precise estimate of bonnet macaque in the wild is available, it is believed that the population of macaques in many regions has increased many-fold. It is a fact that populations rapidly increase in the surrounding region of human habitations.

According to the official and media reports, 20 States/UTs have reported significant crop damages due to monkey attacks. Monkeys relish a wide range of food, including roots, shoots, leaves, fruits, and even grasses. The problem of monkey menace is complex and needs a coordinated effort by the Agriculture, Horticulture, Animal Husbandry, and Forest Departments. The strategies recommended are as follows:

1. Integrated Farming System approach under which the following activities are required to be adopted.

a. Cultivation of crops not damaged by monkeys; crops like ginger, turmeric, Colocasia, yam, Marigold, lime, lemon, lemongrass etc.

b. Promotion of dairy farming, which requires area under fodder cultivation and creation of fodder banks.

c. Enriching habitats in deep forest areas so that sufficient food remains available to them thereby containing the monkey population within the deep forest by raising natural wild fruit plants.

2. In addition to the above measures, it is recommended to forcibly translocate problematic monkeys to specially created monkey shelters. It is proposed to create four monkey shelters in high conflict regions in the State that can accommodate 250 monkeys (Bonnet macaque) each.

Monkeys should be captured using appropriate methods by hiring professional monkey trappers and the animals of both sexes should be sterilized before releasing into the shelter. Initially, the monkey shelters should be created in locations like Wayanad and Malappuram districts where the menace is severe.

A 15-meter-tall chain link fence lined with fibre sheets on the inner surface could be erected to create an enclosure to prevent monkeys from climbing up. Food including fruits and vegetables is to be purchased from the market and regularly supplied to the shelters. This model tried elsewhere can be replicated here. The use of sound blasters with human presence scares the monkeys away from crop fields. Placement of some olfactory repellents also reduces the monkey problem, especially on coconut crowns.

Dealing with peafowl and grey-headed swamphen

The peafowl population is increasing in Kerala, and they are spreading to newer areas. The

possible reason attributed is the absence of its natural predators in the ecosystem. However, there is no scientific data available to correlate increased population status and its damage to the ecosystem. In this proposal, it is suggested to study the population status and impact on crop damage based on the scientific assessments/studies. Correlation analysis between weather parameters and incidence of grey-headed swamphen in the agricultural ecosystem in the central plains of Kerala during the period from 2016 to 2021 reveals that there is a significant positive correlation between temperature and the presence of grey-headed swamp hen.

In some locations, the grey-headed swamp hen has been reported to be causing heavy damage to rice cultivation. The uncultivated lands in the adjoining areas are used by moorhens to roost. This problem can be solved to a great extent by clearing the vegetation viz., Salvinia and water hyacinth to discourage roosting especially in waterlogged areas. The LSG should take lead in utilizing the service of NREGS to address the issue effectively.

GENERAL RECOMMENDATIONS

The present mitigation measures do not consider the landscape as a unit. Thus, the unit for mitigation is either the Forest Divisions or even Ranges. Landscapes should be identified based on the contiguity of geographical areas/forests. Conflict areas requiring interventions should be prioritised based on selected parameters such as the severity of conflict incidents, number of affected people, location of the areas, and socioeconomic status of the affected people.

General

- Mitigation measures should be selected based on the cost-effectiveness, nature of the terrain, soil types, and social acceptability, as well as the movement path of animals.
- The details of selected areas, the planned mitigation measures, and the implementation procedures should be publicized through websites and preliminary discussions could be held with the elected representatives of the people. The general public could also be requested to comment on the plan. However, the decisions should never be subjective and not influenced by considerations other than technical. This comprehensive landscape-level plan should be the basis for further activities.
- Alternatively, the whole process of prioritisation, selection of sites, and the mitigation techniques could be done by a team of experts and presented before the Forest Officials for improvement.
- Permission from the Forest Department should be made mandatory for any type of physical barriers planned by private people or departments. This is to ensure that the natural movement of animals is not hampered.
- Priority should be given to voluntary rehabilitation of people, with priority for Wayanad, to make available inviolate areas for wildlife and reduce the chances of a conflict. This would be the ultimate solution to the problem.
- Keep the farm boundaries clear of palatable vegetation and maintain a buffer zone between farmland and forest: As the foraging behaviour of all ungulates is strongly influenced by the potential escape cover existing along the periphery of the farmland,

regular clearance of the palatable vegetation would be a viable strategy.

- The State should utilise the provisions of the Wildlife Protection Act to deal with the issue of wild boar, which is considered the major problematic animal by farmers. Any such actions should be monitored, and corrective measures are taken.
- Early systematic burning of grasslands facilitates the provisioning of nutrition-rich new flush attracting all herbivores including elephants. Once the grass is dry, its nutrition value is very low. Moreover, tribal communities use this as a traditional practice to prevent herbivores from feeding on their crops.
- Wherever there is the likelihood of people venturing out before daylight, red light signals can be provided in prominent places to alert people about the presence of elephants because many deaths happen when people accidentally encounter elephants without being aware of their presence.
- The process of acquiring lands in identified corridors needs to be expedited. Roads in hilly terrain going through elephant habitats often block normal elephant passages due to sharp cutting and filling works. This leads to aggression towards vehicles and people. Such areas need to be identified and made passable for elephants by easing of gradient and creating ramps. Roads through Chinnar, Nadugani, Nelliampathy, and Pampa are examples.
- The habitat improvement activities are a continuous programme that needs to be strengthened and standardized by the Forest department. Large animals and herbivores prefer open spaces inside forests and such areas should be allowed to remain as such. Species like Grewia tiliaefolia can be planted extensively in areas where invasive species are removed.
- After identifying the problematic animal, if necessary, the animal should be captured using specific methods based on the existing guidelines.

Solar-powered Fences

- Failure of Solar Power Fences is mostly because of problems of low-quality materials, improper implementation, and lack of maintenance and care by the beneficiaries. A local Panchayath-level Committee should oversee the implementation of the mitigation measures. They should ensure the quality of the materials used and also the quality of the work.
- If solar power is adopted, it is suggested to have at least one field watcher/5 km for guarding at night and maintaining fences. Discussions may be held with Panchayaths to meet the related expenditure.
- The suggestions from experienced Foresters indicate that there should be an energiser and a battery for every 1.5 km of solar power fence. This may be evaluated technically.
- Maintenance for a fixed period should be ensured while selecting the agency and in the case of solar power fence, the size of the posts, and specifications of the energisers and battery should be tested by the Panchayath-level committee.
- Arrangements should be made to address any technical problem, especially with solar power fences within one or two days.
- There should be periodic performance auditing by the staff for immediate repair

if required. The Panchayath Members also may be involved in monthly reviews of performance. A Committee may be constituted to monitor and evaluate the performance at least once in six months. The field staff and Division level officials should also monitor the measures taken periodically.

- It is recommended that the maintenance of solar power fences / hanging solar power fences should be included as an item in the NREGS scheme to ensure continuous availability of funds and generation of employment to local communities.
- Stone wall construction should be avoided because of the high cost and environment-related reasons. The higher cost and the recent reports of negative impact on elephants indicate that the rail fence should also be avoided. A crash guard fence is also very expensive and there is every likelihood of elephants getting hung on the ropes just like it happens on rail fences. Likewise, steel structures with spikes should not be allowed.

Capacity Building

- Several Primary Response Teams (PRT) should be created under JJS. PRT would consist of local people with considerable knowledge about WL, locality, and animal behaviour. The front-line forest staff and volunteers could be a part of this. The area of operation of PRTs would be very small and within easy reach of its members. The task assigned to PRTs would be to deal with conflict issues before they get magnified into major conflicts. They will be the first to intervene in conflict situations as first responders to deal with wild animals by managing conflict situations immediately, thereby effectively averting conflicts getting magnified. PRTs should be formed in consultation with Grama Panchayaths. The team should consist of a minimum of 10 persons under the leadership of the frontline staff of the Forest Department. They should be well equipped and trained. NGOs and NGIs who wish to collaborate with PRTs can be co-opted after proper scrutiny. It is proposed to have at least one PRT for each locality in the Ranges having high HWC issues.
- More JJS has to be formed in conflict areas, especially human-elephant conflict areas.
- More RRTs have to be formed. The RRTs should be strengthened and well equipped. The manpower should be increased so that there are enough people to be engaged on rotation.

Involving Other Departments

- Currently, several Government Departments extend financial support to mitigation measures. As can be seen in different areas, District Panchayaths, Grama Panchayaths, and Tribal departments are involved in the activities. There is not much coordination between departments and the decisions are subjective. The Forest Department should present the details of the report having prioritisation and mitigation actions before the concerned line departments and seek assistance. At any cost, the line departments should not be allowed to implement any mitigation activity without the approval of the Chief Wildlife Warden.
- The ex-gratia for damage to crops due to wild animals shall be made the responsibility of the Department of Agriculture.

- The full responsibility of managing conflicts in Exclusion Zones should be handed over to the Panchayaths. National Board for Wildlife has already given the green light for empowering Grama Panchayaths in dealing with problematic wild animals as per Section 11 (1) (b) of Wildlife (Protection) Act.
- Garbage is known to attract wildlife, especially the wild boar and monkeys, and when these dumps are on the periphery or inside the village/town it creates the potential for accidental encounters between people and wildlife. Unmanaged garbage may also habituate elephants to moving and foraging in human-use areas and as a consequence, it may result in high levels of conflict. Considering these, Town Municipalities and Village Panchayaths bordering elephant habitats should be engaged by Forest Department to ensure that there is sustainable and ecologically sound garbage disposal and periodic inspection of the forest perimeter near villages/towns should be undertaken to ensure that poor disposal of garbage is detected and brought to the notice of the local authorities.
- In places where some elephants have become habituated to foraging inside the boundary of villages and towns in search of garbage, 'aversion conditioning' should be tested if they do not stop this behaviour even after the garbage is no longer available.
- In addition, sensitizing local communities and truck drivers on elephant behaviour and movement patterns, signage along roads and railway, proper waste disposal facilities, routine inspection of roads and railway lines should be done by Forest Department and associated stakeholders to mitigate such problems.

Source of Funding

- The funding for HWC mitigation must be mainly from the Government. In situations of shortages of the fund, the KFD through Jan Jagratha Samithis shall take appropriate measures to channel funds from MP and MLA LAD funds and funds from District, Block, and Grama Panchayaths.

Ex-gratia and Financial Incentives

- The present rates fixed for 'compensating' the loss are to be suitably enhanced. The insurance scheme by the Agricultural Department for damages to crops by wildlife needs to be continued.
- Introducing appropriate Crop Insurance Schemes for the high-risk croplands to mop up the financial loss in the event of a crop raid is proposed. Appropriate policy decisions may be taken at the government level for deciding the percentage of premium to be borne by the State.
- Periodic revision of the rates of ex-gratia to be paid for wildlife damage should be done by Agriculture Department in consultation with the Forest Department.
- Crops such as mango ginger (Curcuma amada Roxb.), which wild pigs detest, have been profitably planted in fields bordering forests. Financial incentives at the rate of Rs.1000/- per standard bed can be given to farmers who adopt this practice. Another crop that can be tried in suitable areas is lemon. It is financially attractive and not raided by any animals.

- The appropriate subsidy can be given to farmers in the forest fringes who are facing wild pig problems in their agricultural lands for Wild pig-proof fences to protect the farmlands in the forest fringes.
- If the responsibility of maintenance of the barrier is on the beneficiaries, claims for ex-gratia for crop/property damage from areas where human habitations are completely secured should not be entertained.
- In case of a location/person having received ex-gratia more than four times, no ex-gratia payment shall be made.

Capture of Problematic Animal

- Wild animals should not be captured physically unless required. Decisions about whether to release the animal in another area or retain it in captivity should be based on a scientific understanding of animal behaviour and experience with earlier management actions. A zone of exclusion has to be delineated as an exclusion zone that is outside the normal home ranges of animals like elephants and tigers.
- Regarding capture and relocation, the veterinarians need to be trained in ballistics and long-range shooting since a successful chemical restraint is dependent on several factors like drug used, dosage, body score, site of injection, knowledge of ballistics, and the landscape. Negligence of any of these factors may even cost the life of the animal. At present, the competence level of vets in handling sophisticated long-range tranquilizing equipment in open fields is not adept for critical operations. Hence training in international centers should be imparted to forest veterinary officers in the use of long-range rifles for field operations.
- The wild pig permanently housed in areas away from the forest should be captured using drop-door cages. The pigs so captured should be either culled or released in tiger-bearing forests to augment the prey base.
- Rescue Centres have to be established in selected areas for dealing with injured carnivores that are to be captured and treated and subsequently returned to the wild.
- The elephants that are habitual in raiding crops and attacking people causing severe loss of life and property should be captured and kept in captivity for the rest of their life. The animals that are fit to be trained as Kumkies should be shifted to the Elephant Camp located at Wayanad WLS for training. Other elephants can be kept in elephant rehabilitation centers available in the State.
- Each incidence of accident leading to human death should be analysed in detail for understanding the situations leading to the death and make corrections in management.
- In the case of tigers that have become man-eaters or habitual in livestock depredation, the protocols issued by NTCA shall strictly be adopted. The Center for tiger rehabilitation already developed in Wayanad shall be replicated in Thrissur and maintained.
  In the case of tigers and leopards, the guidelines issued by the Ministry of Environment, Forests, and Climate Change should be followed.
- The scope of Assistant Forest Veterinary Officers has to be broadened much further by using AVO as a component of RRT/PRT institutions and also utilizing his professional knowledge in interacting with fringe area communities.

13. Extension Activities

- Form a' citizen conservationist' group of enthusiastic youth among the local community. They can conduct field patrolling on a regular rotation basis to keep away all wild animals from the farmland.
- Periodic Interaction Meetings with elected people's representatives should be organised for better public relations. Workshops for media persons could also be organised.
- A Forest Department staff, who is knowledgeable and good at communication should be selected and entrusted to deal with the media at the time of conflict.
- It has been argued that a person's social network, social standing, gender, financial stability, beliefs, and values contribute to their decisions about how to engage with their environment. The attitude and behaviour of the people influence their relationship with the local authorities. It is important to cultivate trust with the local people while maintaining authority. This has been evident at least in Wayanad while managing conflict situations. Though skilled in dealing with illegal activities, the Forest department staffs need training in gaining the trust of the people. In-service training for the staff may be thought of. This could also be a part of the syllabus in Forest schools.
- The data on population ('wildlife census') collected periodically may be analysed to see the trend in the population at the regional and landscape level. Appropriate population management following the provisions of the Act could be considered based on the information.
- The impact of tourism on wildlife, especially in conflict areas needs to be evaluated.
- The Forest department is to develop an increasing presence on social media to counter fake reports and scaremongering on conflict issues.
- New studies need to be initiated focussing on communities and their perception and tolerance levels rather than focusing on animals alone.

2. Physical and Financial Requirements for Mitigation of Human-Wildlife Conflict in the State

PHYSICAL BARRIERS

The degree of conflict between humans and wildlife greatly varies with the regions in the State. In areas where a particular population of wildlife species (which share the resources with humans) is higher along with a high density of human population, the conflict situations due to these species are also higher. The preventive measures existing in each Division are listed in Annexures 2 and 3. The proposed preventive structures required for the next 5 years division wise is given in Annexures 4 & 5.

Elephant-proof Trenches

The division-wise requirement of EPT is given in Annexure 4, and the circle-wise requirement is summarized in Table 9.

Table 9: Summary of the requirement for Elephant-proof Trench

Circle	Elephant-proof Trench (Km.)
Agasthyavanam Biological Park	2.00
Central Circle, Thrissur	25.00
Eastern Circle, Palakkad	15.00
Field Director (PT), Kottayam	10.00
High Range Circle, Kottayam	20.00
Northern Circle, Kannur	20.00
Southern Circle, Kollam	30.00
Wildlife Circle, Palakkad	10.00
Grand Total	132.00

Solar-powered Fence:

The division-wise requirement of the Solar Power Fence is given in Annexure 4 and the circle-wise requirement is summarized in Table 10.

Hanging Solar-powered Fence:

The division-wise requirement of this component is given in Annexure 4 and the circle-wise requirement is summarized in Table 11.

Bio-Fence

The division-wise requirement of this component is given in Annexure 4 and the circle-wise requirement is summarized in Table 12.

Table 10: Summary of the requirement of SPF

Circle	SPF (Km.)
Agasthyavanam Biological Park	10.00
Central Circle, Thrissur	80.00
Eastern Circle, Palakkad	125.00
Field Director (PT), Kottayam	45.00
High Range Circle, Kottayam	125.00
Northern Circle, Kannur	125.00
Southern Circle, Kollam	150.00
Wildlife Circle, Palakkad	20.00
Grand Total	680.00

Table 11: Summary of the requirement of HSPF

Circle	HSPF (Km.)
Agasthyavanam Biological Park	5.00
Central Circle, Thrissur	80.00
Eastern Circle, Palakkad	125.00
Field Director (PT), Kottayam	15.00
High Range Circle, Kottayam	65.00
Northern Circle, Kannur	185.00
Southern Circle, Kollam	85.00
Wildlife Circle, Palakkad	125.00
Grand Total	685.00

Table 12: Summary of the requirement of bio-fence

Circle	Bio-fence (Km.)
Agasthyavanam Biological Park	10.00
Central Circle, Thrissur	10.00
Eastern Circle, Palakkad	10.00
Field Director (PT), Kottayam	10.00
High Range Circle, Kottayam	10.00
Northern Circle, Kannur	10.00
Southern Circle, Kollam	15.00
Wildlife Circle, Palakkad	10.00
Grand Total	85.00

Measures to avoid the interface between People and Animals

Early Warning System
The circle-wise requirement of this component is given in Annexure 4 and summarized in Table 13.

Table 13: Summary of the requirement of the early warning system

Circle	Early Warning System (No.)
Agasthyavanam Biological Park	1
Central Circle, Thrissur	4
Eastern Circle, Palakkad	5
Field Director (PT), Kottayam	2
High Range Circle, Kottayam	5
Northern Circle, Kannur	5
Southern Circle, Kollam	5
Wildlife Circle, Palakkad	5
Grand Total	32

Use of Drones:

The circle-wise requirement of this component is given in Annexure 4 and summarized in Table 14.

Table 14: Requirement of Drones for scaring/monitoring of animals

Circle	Drones (No.)
Agasthyavanam Biological Park	2
Central Circle, Thrissur	4
Eastern Circle, Palakkad	5
Field Director (PT), Kottayam	4
High Range Circle, Kottayam	5
Northern Circle, Kannur	5
Southern Circle, Kollam	6
Wildlife Circle, Palakkad	5
Grand Total	36

Radio-collaring of problematic animals: The problematic animals and habitual crop raiders may be captured with the permission of the Chief Wildlife Warden and radio-collared for monitoring their movement. This requires identifying and collaring individual problem animals, receiving GPS locations via satellite or GSM technology, mapping the location on GIS maps, and informing the people who are at risk. This would generate information on

temporal and spatial patterns of crop-raiding, possible contributory factors, the effectiveness of preventive methods, etc. Moreover, if elephants are not residents, only seasonal visitors, theoretically it is possible to monitor the movements of individual crop raiders through GPS satellite radio collars and warn villagers of their presence. Hence, it is proposed to procure eight radio-telemetries with satellite subscriptions and hand-held receivers. The data thus collected using the satellite will appropriately be collated and analysed for devising strategies to mitigate HWC in the State. The procured radio collars may be provided to the Tiger Conservation Foundations (TCFs) existing in the State and utilized as and when required. The necessary financial requirement for this component is included in the budget provision.

Translocation of problematic animals: Capture and translocation of problem animals had been suggested on many occasions for species that create havoc in human habitations. Though experience elsewhere suggests that the translocation does not always help as the released ones return to the site of capture or recreate the same problem at their release site, this method is suggested as a last alternate solution only for animals that create serious problems in the human habitations. Translocation of elephants especially in the Wayanad region proved to reduce the problem as the visits of the identified habitual crop raiders to the affected area have come down considerably. In the case of tigers and leopards, the capture of problem animals in cages used to be a regular practice in the state. Monkeys, especially the Bonnet macaque is considered to be a problem even in villages far away from forested areas. Hence, this method was suggested in many Forest Divisions in the State. For capture and translocation of animals especially the endangered species, various guidelines issued by both State and Central Governments shall strictly be adopted. Necessary financial provision is proposed in this project to deal with the translocation of problem animals.

Infrastructure and Human Resource Development.

Establishment and strengthening of Rapid Response Team (RRT)

In case of any conflict quick response from the Department is very important for preventing loss of human life or damage to property. To ensure this KFD has established a Rapid Response Team in high conflict zone areas. Though several requests were made to increase the number of RRTs, presently only in eight locations approved RRTs are in place in different locations of the State (Annexure 6). It is very imperative to have full-fledged RRTs in all the conflict areas to address the issues immediately. The requirements of RRTs in various Forest Divisions are given in Annexure 5. A total of 25 new RRTs in the following conflict Divisions/locations (Table 15) shall be created and strengthened with all infrastructures including vehicles, equipment, etc. to meet the emergencies in addition to the existing RRTs (eight nos.).

Each newly created RRTs shall be equipped with a vehicle (a total of 25 vehicles – TATA 407 / Canter with 4x4 option - @ 10 lakhs/vehicle), that can also be used as an animal ambulance whenever required. Equipment like rope ladder, net, snake tongs and handling kit, Aska light, torch, handheld cracker projector, stretcher, animal cages, uniform for the staff, lathi, helmet and body armor, spear for tracking tiger, tranquilizer guns with sufficient quantity of tranquilizers/drugs, foothold traps, rifles, camera traps, range finders, binoculars,

walkie talkies, GPS, camping and field gears for the staff dealing with conflict situations, spotlights, night vision equipment, head-mounted lights, nets, etc. (33 kits for RRTs @ Rs. 15 lakhs per kit) shall be procured for strengthening the newly proposed as well as existing RRTs in the State. Satellite phones should be provided to selected RRTs working in very remote areas. Provision for procuring 5 satellite phones has been kept in this proposal.

Table 15: Requirement of new RRTs

Sl.No	Name of the Circle	Name of the Divisions require the establishment of new RRT
1	Southern Circle	Achencoil
2		Thenmala
3		Konni
4		Kottayam
5		Kothamangalam
6	High Range Circle	Munnar
7		Mankulam
8		Marayoor
9		Malayattoor
10	Central Circle	Thrissur
11		Chalakkudy
12		Vazhachal
13		Nemmara
14	Eastern Circle	Mannarkad
15		Nilambur South
16		Kasargod (at Kallar in Kanjangad Range)
17	Northern Circle	Wayanad North
18		Wayanad South
19		Idukki
20	FD, Kottayam	Munnar WL
21		Periyar East
22		Periyar West
23		Aralam
24	WL, Palakkad	Silent Valley
25		Peechi

Formation of Primary Response Teams (PRTs)

The a~~nimals that come into~~ conflict with humans need to be managed tactfully without

incurring any damage on either side. Numerous nonlethal approaches are available to reduce and avoid conflict, and these approaches are often preferable for species of conservation concern. The approaches include methods to drive away problem animals; separate problematic animals from people and livestock and ensure safe passage; and use chemical immobilization and translocation. RRTs will cater to the needs of the entire Division. However, to address the recurrent issue of conflict, it has been proposed to form PRTs either at Range or Forest Station level and function under JSS.

The proposal is to constitute special teams with personnel knowing locality and animal behaviour to keep away the wild animals as well as to manage the conflict situations immediately. The presence of a team of people in the uniform led by well-trained local frontline staff adequately equipped to deal with conflict situations often pacifies agitated people and reassures them. The members of the team (4-6 staff) should be handpicked from the existing staff. It is proposed to select and engage ten numbers of watchers from local areas to support the team. They will be given training on various aspects of wildlife management including the biology and behaviour of animals involved in the conflict, public relations, and also on handling difficult situations. It is proposed to have at least one conflict management team for each Range having high HWC issues. The number of teams required circle wise listed below in Table 16.

Table 16: Circle-wise summary of the requirement of PRTs

Circle	Divisions required PRT (No.)
Agasthyavanam Biological Park	1
Central Circle, Thrissur	8
Eastern Circle, Palakkad	10
Field Director (PT), Kottayam	4
High Range Circle, Kottayam	10
Northern Circle, Kannur	10
Southern Circle, Kollam	12
Wildlife Circle, Palakkad	5
Grand Total	60

Vehicles (4X4 DOUBLE CABIN PICK-UP vehicle @ 8 lakhs/vehicle) shall be provided to every PRT apart from equipment such as rifles, spotlights, cracker projector guns, etc (60 kits @ Rs. 5 lakhs per kit).

In addition, Forest Stations shall be formed wherever required to improve the staff strength as well as basic facilities for effective management of forests and wildlife.

Establishment of Animal Rescue Centres (ARC): Injured carnivores that are unable to hunt in the wild, attack livestock, and even people. Such animals are to be captured and treated and subsequently returned to the wild. These centres can attend to a wide range of wildlife emergencies resulting from natural or anthropogenic causes. The centres can

be maintained under the supervision of Asst. Forest Veterinary Officers and supported by animal keepers who are trained to handle wild animals. The ARCs are to be established with essential veterinary infrastructure including an examination clinic, a surgery theatre, and a preliminary disease screening laboratory. It will also have spacious holding shelters for various species of birds and animals. The centres will help in providing medical aid to stranded, displaced, or distressed animals also. Hence animal rescue centres are proposed in centralized locations of various forest circles in Kerala. The Division-wise requirement of ARC is given in Annexure 5 total of 15 such ARCs are proposed as summarized circle-wise in Table 17

Table 17: Circle-wise summary of the requirement of ARC

Circle	ARC (No.)
Central Circle, Thrissur	2
Eastern Circle, Palakkad	2
Field Director (PT), Kottayam	2
High Range Circle, Kottayam	2
Northern Circle, Kannur	3
Southern Circle, Kollam	3
Wildlife Circle, Palakkad	1
Grand Total	15

Dealing with highly problematic and dangerous animals: The elephants that habitually raid crops and attack people causing severe loss of life and property should be captured and kept in captivity for the rest of their life. The animals that are fit to be trained as Kumkies should be shifted to the Elephant Camp located at Wayanad WLS for training. Other elephants can be kept in elephant rehabilitation centres available in the State (e.g. Kottur). Additional facilities should be created in the elephant camps at Wayanad and Konni for the keeping and training of such elephants.

In the case of tigers that have become man-eaters or habitual in livestock depredation, the protocols issued by NTCA shall strictly be adopted. The centre for tiger rehabilitation already developed in Wayanad shall be replicated in Thrissur and maintained.

Construction of watchtowers. To keep a watch on intruding wild animals into human habitations, tall standing watchtowers are proposed to be constructed in selected locations as proposed by the field officers. A total of seven watchtowers (@ Rs. 30 lakhs per structure) are proposed to be constructed.

Insurances And Ex-Gratia

Ex-gratia for damages to crops. Providing ex-gratia is an approach to provide immediate relief to people adversely affected by wild animals where they are compensated for the loss they sustained. However, payment of ex-gratia is unable to decrease the level of the problem and the cause of the problem is not being addressed. Timely payment of ex-gratia

is also crucial to get conservation support from the people. However, this component is very crucial to keep up the tolerance of the people and to ensure their cooperation for coexistence. In case of livestock-lifting incidents (both injury/kill) happening within human habitation, procedures for providing ex-gratia shall immediately be initiated. In case of ex-gratia for livestock loss, the body of the attacked animal should be photographed and certified by a veterinary doctor. But if the kill is retrieved for verification, the chances of immediate and subsequent attacks are higher. Presently the forest divisions are not able to release all the claims of ex-gratia due to a shortage of funds. The existing rate of ex-gratia for various crops damaged by wild animals has not been revised for a long time and many farmers and peoples' representatives have demanded an urgent revision of the rates. Hence appropriate actions shall be initiated by KFD to revise the existing rates of ex-gratia for crops damaged by wild animals and provide sufficient ex-gratia to farmers. Based on the average amount claimed during the previous years and keeping because of the expected revision of rates of ex-gratia, a lumpsum provision of Rs.15 crores/year is proposed to be set aside for payment of ex-gratia. The rates for damage to various crops should be appropriately revised to compensate for the actual damages.

Ex-gratia payment for loss of life/injury to human life: In some unfortunate events, human lives are lost by wildlife attacks. The staff of the forest department who are deployed for managing HWC are also facing serious risks to their lives. There should be a provision for immediate disbursement of ex-gratia to the victims. The usual financial procedure often causes inordinate delay in the release of the money. Ensuring the availability of sufficient funds with all the forest divisions all the time will minimize the delay in payment of ex-gratia.

Life and Health Insurance for the field staff. The staff engaged in RRTs and PRTs often face dangerous situations while trying to drive the animals away from human habitations and croplands. These frontline staff should be provided a life insurance cover of at least 25 lakh rupees along with provision for cashless treatment in all hospitals. This will boost the morale of the staff and encourage them to work more efficiently. Adequate provision for the payment of premium has been kept in this proposal.

Crop Insurance for wildlife damage. Introducing appropriate Crop Insurance Schemes for these high-risk croplands to mop up the financial losses in the event of a crop raid is proposed. Appropriate policy decisions may be taken at the government level for deciding the percentage of premium to be borne by the State.

Involving Panchayaths and Line Departments in Assisting HWC Management

- Periodic revision of the rates of ex-gratia to be paid for wildlife damage should be done by Agriculture Department in consultation with the Forest Department. The ex-gratia for damage to crops due to wild animals shall be made the responsibility of the Department of Agriculture.
- It is strongly recommended that the maintenance of solar power fences/ hanging solar power fences should be included as an item in the NREGS scheme to ensure continuous availability of funds and generation of employment to local communities.
- In situations of fund shortages, the KFD through Jan Jagratha Samithis shall take

appropriate measures to channel funds from MP and MLA LAD funds and funds from District, Block and Gram Panchayath for various HWC mitigation measures.

Strengthening People's Participation in HWC Mitigation

- Strengthening Jan Jagratha Samithis. The Jan Jagratha Samithis should be developed as an important body at the level of a Forest Range sharing boundary with Panchayath having high HWC issues. To take appropriate decisions, the Committee should often meet, review the HWC issues and take appropriate actions.
- Involvement of VSSs/EDCs. The involvement of EDCs/VSSs members is also included in the management of high HWC areas. Necessary support from Forest Development Agencies and Tiger Conservation Foundations shall be provided for creating awareness and sensitization of various stakeholders on HWC issues and providing small financial assistance during emergencies.

Habitat Quality Improvement Activities.

Vista clearance. Clearing vegetation along the roadside and around human habitation is being practiced in many forest divisions to have a clear view of areas used by the local community and to avoid HWC. Many incidents of human deaths in forest areas have taken place in the surrounding areas of human habitations with thick vegetative cover. Hence, this component is proposed to be implemented as one of the tools to reduce the HWC, especially in areas of frequent human visitations including regular trek routes in which the forest staff move, public roads passing through the forest, and roads leading to human settlements, etc. The requirement at the divisionlevel during the initial 5 years is given in Annexure 5 and summarized circle-wise in Table 18.

Table 18. Circle-wise summary of the requirement of vista-clearance

Circle	Vista-clearance (Ha.)
Agasthyavanam Biological Park	50.00
Central Circle, Thrissur	250.00
Eastern Circle, Palakkad	250.00
Field Director (PT), Kottayam	350.00
High Range Circle, Kottayam	200.00
Northern Circle, Kannur	250.00
Southern Circle, Kollam	250.00
Wildlife Circle, Palakkad	400.00
Grand Total	2000.00

Maintenance of waterholes. Since surface water repositories play a critical role in the movement of animals in Protected Areas and wildlife corridors, particularly in the hot months, it is essential to maintain waterholes throughout the forest. The waterholes maximize water retention in the soil and are crucial to ecosystem health. Thirst can be one of the major causes of wildlife mortality in the summer months, either directly or by

driving animals to stray out of their natural habitat and into conflict with humans. While the seasonal depletion of water bodies is a natural phenomenon in certain landscapes, it has, in recent years, been accentuated by climate change and anthropogenic alterations to habitat. Hence this component is highly essential for managing/containing wildlife in their natural habitat and mitigating HWC. The requirement maintenance of waterholes during the initial 5 years in division-wise is given in Annexure 5 and summarized circle-wise in Table 19.

Table 19: Circle-wise summary of waterholes to be maintained

Circle	Maintenance of waterholes (No.)
Agasthyavanam Biological Park	25
Central Circle, Thrissur	55
Eastern Circle, Palakkad	100
Field Director (PT), Kottayam	200
High Range Circle, Kottayam	50
Northern Circle, Kannur	50
Southern Circle, Kollam	150
Wildlife Circle, Palakkad	220
Grand Total	850

Creation of brushwood check dams. A check dam is a barrier constructed across waterways such as open channels, or streams. Check dams retain water behind them that helps the surrounding soil to remain wet for a long time after the monsoon season is over. Check dam can either be a temporary or a permanent structure, which is built to retain and maintain the desired groundwater level.

Check dams to maintain high soil moisture levels to minimize fire in the forest areas from occurring and spreading especially during the prolonged dry and hot season to conserve as much water as possible during the post-monsoon season that would help to recharge groundwater. Brushwood check dams are proposed wherever it is feasible. The number of proposed brushwood check dams during the initial five years in division-wise is given in Annexure 5 and summarized circle-wise in Table 20.

Vayal management. The 'Low altitude marshy grasslands' (known as vayals) are a unique feature in most of the forest areas in Kerala. This is a crucial ecosystem or microhabitat to support the herbivore populations in the forest areas. Grasses and sedges are the important components that form a high-density feeding ground for the herbivores during the dry season. These vayals contain ample food for herbivores in addition to water that flows through these microhabitats. Most of these vayals are presently threatened by the invasion of exotic species and woody species. To maintain the contain the herbivore population within the forest, these vayals need to be maintained. The requirements of vayal management during the initial 5 years in division wise are given in Annexure 5 and summarized circle-wise in Table 21.

Table 20: Circle-wise summary of the creation of brushwood check dams

Circle	Creation of brushwood check dams (No.)
Agasthyavanam Biological Park	50
Central Circle, Thrissur	250
Eastern Circle, Palakkad	250
Field Director (PT), Kottayam	350
High Range Circle, Kottayam	250
Northern Circle, Kannur	250
Southern Circle, Kollam	250
Wildlife Circle, Palakkad	350
Grand Total	2000

Table 21: Circle-wise summary of vayal management

Circle	Vayal management (Ha.)
Agasthyavanam Biological Park	20
Central Circle, Thrissur	80
Eastern Circle, Palakkad	100
Field Director (PT), Kottayam	200
High Range Circle, Kottayam	150
Northern Circle, Kannur	100
Southern Circle, Kollam	100
Wildlife Circle, Palakkad	500
Grand Total	1250

Removal of invasive alien species. The invasive species such as Senna spectabilis, Lantana camera, Eupatorium, Mikenia, and Parthenium are not palatable for herbivores and their invasion of the natural forest areas brings down the food base of the herbivores. Any setback to the herbivore population owing to the non-availability of fodder will, in the long run, have a reflection on the carnivore population as well. More than reducing the availability of natural fodder for wild animals, the weeds compete directly for light, nutrients, moisture, and space with the native forage of wildlife. Hence periodic removal of these exotic species is highly crucial for the maintenance and management of forests with sufficient fodder for the herbivores. Hence this component of removal of exotic weeds is proposed to be implemented in most of the forest divisions in Kerala (Annexure 5) and the circle-wise requirement during the initial 5 years is given in Table 22.

Table 22: Circle-wise summary of exotic weed management

Circle	Exotic weed management (Ha.)
Agasthyavanam Biological Park	50
Central Circle, Thrissur	500
Eastern Circle, Palakkad	400
Field Director (PT), Kottayam	400
High Range Circle, Kottayam	400
Northern Circle, Kannur	500
Southern Circle, Kollam	400
Wildlife Circle, Palakkad	1350
Grand Total	4000.00

Enrichment planting. Fruit species such as jack fruit, mango, and other preferred indigenous edible species shall be planted in the degraded forest areas to contain the crop-raiding animal species. It is proposed to treat 5000 Ha. In this manner.

Eco-restoration of industrial plantations of Exotic Species. There is about 30,000 ha of industrial plantations of exotic species of Eucalyptus, Acacia auriculiformis, Acacia mangium, and Black wattle. Planting of these species is dispensed within 2018. These plantation areas are to be restored to their natural condition in a phased manner after the harvest so that the habitat quality will be improved for catering to the needs of wildlife. It is proposed to commence the eco-restoration works in at least 5000 ha of these exotic plantations during the initial 5 years.

Training, Awareness Creation, and Research

Training support. Training for staff involved in human-wildlife conflict mitigation measures is essential for the effective management of human-wildlife conflicts. The training may include sessions on social, biological, and veterinary aspects related to human-wildlife conflict. The training must include both theory and practical sessions. The theory sessions such as causes and consequences of conflict, mitigation strategies, introduction to chemical immobilization of wild animals and the equipment used therein, monitoring of wildlife signs, documentation and use of GPS, camera trapping, and the deployment of trap cages, etc. Sociological elements would include community engagement in conflict-affected areas, case studies, and the role of the media in conflict mitigation. The staff or the members in the training program can share their understanding of conflict, its causes, and possible solutions, and discuss incidents of human-big cat/larger mammals' conflict that have occurred in various landscapes in the State to have a wider and in-depth knowledge in the subject. In addition, practical sessions including hands-on training with field gear typically used during conflict situations shall be imparted. The mahouts and kavadis shall be imparted with proper training for handling captive elephants/kumkies.

Awareness campaign among the stakeholders. This component is proposed to be outsourced to organizations/institutions having expertise in conducting awareness campaigns on changing cropping patterns, measures to avoid direct interface with wild animals, changes in lifestyles to avoid conflict, and strategies for coexistence to the stakeholder communities.

Decoding of infrared communication of elephants. This component aims at decoding sounds including infra-red sound used for communication by elephants and standardizing the same for early detection and avoidance of conflict situations. The possibility of using the sounds to keep away the elephants from habitation will also be explored. This component is conceived as a pilot project which will be carried out for two years. This component includes provisions for the purchase of equipment and the expenses of engaging a researcher for 24 months. Experiments conducted in Namibia have given promising results in this regard.

Study of migration and crop-raiding patterns of wild animals in selected localities. To understand the dynamics of crop-raiding by various species of animals, a research study has been proposed using modern techniques including radio-tracking the problematic animals. The study is aimed to develop strategies for the management of crop-raiding animals and the adoption of appropriate mitigatory measures.

Maintenance of Existing Infrastructures

In addition, to the above, the existing preventive structures detailed in Table 8 will annually be maintained. The items for maintenance of the Elephant-proof Trench and Solar Electric Fence should be included in the MGNREGA in the DSR of every district. Appropriate directions for the same shall be given to the District Authorities from the State Government.

BUDGET REQUIREMENTS

The total financial budget requirement for the implementation of the above proposal for 10 years is Rs.1150.00 Crore. The physical requirement is projected only for 5 years which will be reviewed after 5 years. The summary of the budget requirement for all proposed activities detailed in this proposal is given in Table 23.

Particulars	Year-wise financial forecast										Total
	1	2	3	4	5	6	7	8	9	10	
Physical barriers	2500	2500	2500	2500	2500	2500	2500	2500	2500	2500	25000
Measures to avoid the interface	1500	1500	1500	1500	1500	1500	1500	1500	1500	1500	15000
Infrastructure and human resource development	1500	1500	1500	1500	1500	1500	1500	1500	1500	1500	15000
Insurances and ex-gratia	2000	2000	2000	2000	2000	2000	2000	2000	2000	2000	20000

Strengthening people's participation in HWC mitigation	100	100	100	100	100	100	100	100	100	100	1000
Habitat quality improvement activities	2200	2200	2200	2200	2200	2200	2200	2200	2200	2200	22000
Training, awareness creation and research	200	200	200	200	200	200	200	200	200	200	2000
Maintenance of existing infrastructures	1500	1500	1500	1500	1500	1500	1500	1500	1500	1500	15000
Grand Total	11500	11500	11500	11500	11500	11500	11500	11500	11500	11500	115000

GOVERNANCE AND POLICY MATTERS

Enhanced levels of Human-Animal Conflicts are experienced across the State. However, the situation is of serious proportion in some locations which require coordinated efforts by various Government Departments / Agencies and the active involvement of local inhabitants. As the issue being faced is the result of a complex interface of multiple factors, its effective management also calls for joint efforts by multiple stakeholders. Important governance interventions required in this regard are the following;

- With the active involvement of all the stakeholders prioritize the locations where the issue is serious. Coordinated efforts by all the stakeholders are to be taken up in these areas on a priority basis.
- Human activities outside the wildlife habitats, especially along the forest periphery have a direct impact on the behaviour of wild animals. This includes activities like quarrying, especially when carried out with the help of explosives. Such activities along the critical areas are to be thoroughly reviewed and regulated/monitored.
- A general policy that discourages the cultivation of crops that attract wild animals, especially along the forest periphery, should be implemented.
- Increasing levels of Human-Animal Conflicts have to be seen in the larger context of global warming and climate change. Therefore, managing these conflicts should become an integral part of the Disaster Management Plan.
- The State / District Disaster Management Agencies need incorporate measures for mitigating human-animal conflict in their works.
- The effectiveness of measures to mitigate Human-Animal Conflict is critically dependent on sensitizing the residents on the need for the peaceful coexistence of humans and wildlife in the given landscape. Efforts in this direction should be actively facilitated by the Local Self Government Institutions (LSGI).
- As the conflict is more intense along the forest periphery, Grama Panchayaths in such areas need to be provided with adequate funds through the Block Panchayath / Jilla

Panchayath.

- The scope and ambit of MGNREGS need to be expanded to include erecting and upkeep of physical barrios and other such activities.
- Payment of ex gratia should be included as an eligible item under the Chief Minister's Disaster Relief Fund.
- Given the significant number of human casualties due to snakebite in areas away from forest periphery, identified hospitals at District / Taluk levels should be facilitated to acquire and stock sufficient quantities of anti-venom.
- Unnecessary crowding in conflict sites which enhances potential risks to both humans and wildlife needs to be prevented using the provisions of CrPC. Forest officers may be authorized to exercise these provisions.
- The Forest Station system with enhanced human resources, vehicles, and equipment should be expanded in all the remaining areas within the next 1-2 years.
- Existing RRTs need to be strengthened and equipped while new RRTs need to be established in the high priority areas.

A joint effort by various Government Departments / Agencies, active participation of LSGI, and sensitisation of residents on the need for peaceful coexistence between human beings and wildlife can go a long way in addressing the present Human-Animal Conflict scenario in the State. The above governance interventions can play a prominent role in these efforts.

3. CONCLUSION

Human-wildlife conflict remains a significant problem for the State. The problem threatens human lives and livelihoods. Current strategies to manage human-wildlife conflict largely focus on either physical separation, or mitigating the problem by domesticating, translocating, or culling problematic animals and/or compensating farmers. While these tools remain important conflict management strategies, the majority appear to be driven by short-term, site-specific factors that often transfer the problems of human-wildlife conflict from one place to another. In this report, the causes and consequences of human-wildlife conflict and current approaches to preventing and mitigating human-wildlife conflict were reviewed. Conflict hotspots were identified and mitigation measures were recommended taking landscape as a unit for future management. The application of ecological, sociological, and geographical knowledge and tools for developing long-term sustainable solutions to this complex problem is highlighted.

The success of any strategy requires inter-departmental cooperation. However, understanding how people living in or near conflict-prone areas use natural resources, and how they make decisions about current and future resource use, remain key to addressing underlying drivers of human-wildlife conflict and their spatial variation. Without this knowledge, the task of resolving human-wildlife conflict and finding a workable solution for these species to coexist with humans is not possible.

Annexure-1

Table A1Existing HWC preventive structures in the forest divisions of Kerala

Circle	SN	Division	Range	Existing Preventing Structures (length in km.)									
				Elephant-proof Trenches	Elephant-proof Walls	Rail Fence	Solar Power Fence	Crash Guard Rope Fence	Hanging fence	Steel structures with spikes	Bio-fencing	Kayyala	Any other structure
SOUTHERN CIRCLE KOLLAM	1	Thiruvananthapuram	Kulathupuzha	2.500			44.650						
			Palode	0.900									
			Paruthippally	6.450			35.300						
	2	Thenmala	Ariyankavu	1.256			53.000					2.500	
			Thenmala				5.000						
	3	Achencoil		1.700			69.397						
	4	Ranni	Ranni		1.540		62.570					0.050	
	5	Punalur	Anchal				12.500						
			Pathanapuram				46.530						
	6	Konni		2.000			131.790					10.000	
HIGH RANGE CIRCLE KOTTAYAM	7	Kothamangalam	Thodupuzha				11.800						
			Kothamangalam				3.000						
			Kaliyar				0.000						
			Mullaringad				0.000						
	8	Munnar		9.250			39.170						
	9	Marayoor		0.800			21.040						
	10	Mankulam		1.300			20.885	1.200					
	11	Kottayam		0.085	0.220		57.591						
CENTRAL CIRCLE THRISSUR	12	Vazhachal		2.620			23.438						
	13	Chalakudy		0.790			77.500					0.284	
	14	Malayattoor		3.700			110.000						
	15	Thrissur		2.300			22.475						
EASTERN CIRCLE PALAKKAD	16	Mannarkkad		7.440			67.450						
	17	Nilambur North		7.203	2.567		126.870						
	18	Nilambur South		1.894	3.850		67.180						
	19	Palakkad		3.810	0.300		93.592	5.700	5.700			2.000	
	20	Nenmara					88.250					0.200	
	21	Kozhikode		2.000			58.500						

Circle	SN	Division	Range	Existing Preventing Structures (length in km.)									
				Elephant-proof Trenches	Elephant-proof Walls	Rail Fence	Solar Power Fence	Crash Guard Rope Fence	Hanging fence	Steel structures with spikes	Bio-fencing	Kayyala	Any other structure
NORTHERN CIRCLE KANNUR	22	Wayanad North		64.772	2.870		241.669						
	23	Wayanad South		140.000	14.937		223.750	25.500	2.000				
	24	Kannur		2.500	10.000		84.940						
	25	Kasaragod		6.200	3.740		87.320		2.000				
AGASTHYAVANAM BIOLOGICAL PARK	26	Thiruvananthapuram WL		3.500			15.000						
	27	Shenthuruni					11.000						
FIELD DIRECTOR (PROJECT TIGER) KOTTAYAM	28	Periyar East		15.760	5.347		15.470					0.090	
	29	Periyar West		0.500	0.053		22.550						
	30	Munnar WL		1.980	0.070		30.920						
	31	Idukki WL	Idukki	0.800			2.500						
			Thattekkad	0.030			16.000						
WILDLIFE CIRCLE PALAKKAD	32	Parambikulam											
	33	Wayanadu (WL)		212.960	5.205	10.000	196.940				0.200		
	34	Silent Valley					24.450						
	35	Peechi WL		0.67			16.85						
	36	Aralam WL	Aralam	3.550	15.565		9.300			1.880			
				511.22	**66.264**	**10.000**	**2348.1369**	**32.400**	**9.700**	**1.880**	**0.200**	**15.124**	**0.000**

Annexure 2

Table A2: Use of other HWC preventive measures in the forest divisions of Kerala

Circle	SN	Division	Presence of Other Preventing Structures	
			Early warning electronic systems	Use of Drones for scaring
Southern Circle Kollam	1	Thiruvananthapuram	Nil	Nil
	2	Thenmala	Nil	Nil
	3	Achencoil	Nil	Nil
	4	Ranni	Nil	Nil
	5	Punalur	Yes	Nil
	6	Konni	Yes	Nil
High Range Circle Kottayam	7	Kothamangalam	Nil	Nil
	8	Munnar	Nil	Nil
	9	Marayoor	Yes	Nil
	10	Mankulam	Nil	Nil
	11	Kottayam	Nil	Nil
Central Circle Thrissur	12	Vazhachal	Nil	Nil
	13	Chalakudy	Nil	Nil
	14	Malayattoor	Nil	Nil
	15	Thrissur	Nil	Nil
	16	Mannarkkad	Nil	Nil
Eastern Circle Palakkad	17	Nilambur North	Nil	Nil
	18	Nilambur South	Nil	Nil
	19	Palakkad	Yes	Nil
	20	Nenmara	Nil	Nil
Northern Circle Kannur	21	Kozhikode	Nil	Yes
	22	Wayanad North	Nil	Nil
	23	Wayanad South	Nil	Nil
	24	Kannur	Nil	Nil
	25	Kasaragod	Nil	Nil

Agasthyavanam Biological Park	26	Thiruvananthapuram WL	Yes	Nil
	27	Shenthuruni	Nil	Nil
	28	Periyar East	Nil	Nil
Field Director (Project Tiger) Kottayam	29	Periyar West	Nil	Nil
	30	Munnar WL	Yes	Nil
	31	Idukki WL	Nil	Nil
	32	Parambikulam	Nil	Nil
	33	Wayanadu (WL)	Yes	Nil
Wildlife Circle Palakkad	34	Silent Valley	Nil	Yes
	35	Peechi WL	Nil	Nil
	36	Aralam WL	Nil	Nil

Annexure 3

Table A3: Proposed preventive structures/measures required by Forest Divisions in Kerala

Circle/Division	Preventive structures/measures required									
	Length in kilometers						Numbers			
	Elephant-proof Trenches	Elephant-proof Wall	Solar Power Fence	Hanging Fence	Bio-fence	Kayyala	Early Warning System	Radio-telemetry	Drone for scaring	Problem elephant relocation
AGASTHYAVANAM BIOLOGICAL PARK	**2.00**		**10.00**	**5.00**	**10.00**		**1**		**2**	
Shenthuruni					5				1	
Thiruvananthapuram WL	2.00		10.00	5.00	5		1		1	
CENTRAL CIRCLE THRISSUR	**25.00**		**80.00**	**80.00**	**10.00**		**4**	**2**	**4**	**1**
Chalakudy			34.00	0.00	2.5		1		1	
Malayattoor	22.50		34.00	75.00	2.5		1	2	1	1
Thrissur			2.00	2.50	2.5		1		1	
Vazhachal	2.50		10.00	2.50	2.5		1		1	
EASTERN CIRCLE PALAKKAD	**15.00**		**125.00**	**125.00**	**10.00**		**5**		**5**	**2**
Mannarkkad	2.00		35.00	35.00	2.0		1		1	
Nenmara	3.00		10.00	10.00	2.0		1		1	
Nilambur North	1.00		50.00	30.00	2.0				1	1
Nilambur South	6.00		20.00	20.00	2.0		2		1	1
Palakkad	3.00		10.00	30.00	2.0		1		1	
FIELD DIRECTOR (PROJECT TIGER) KOTTAYAM	**10.00**		**45.00**	**15.00**	**10.00**		**2**		**4**	**2**
Idukki WL	0.50		4.00	12.50	2.50				1	
Munnar WL	2.50		23.00	0.00	2.50		2		1	
Periyar East	7.00		10.00	0.50	2.50				1	1
Periyar West			8.00	2.00	2.50				1	1
HIGH RANGE CIRCLE KOTTAYAM	**20.00**		**125.00**	**65.00**	**10.00**		**5**		**5**	**1**
Kothamangalam			45.00	10.00	2.0		2		1	
Kottayam	15.00		40.00	20.00	2.0		1		1	1
Mankulam	3.00		15.00	10.00	2.0				1	
Marayoor			15.00	20.00	2.0		1		1	
Munnar	2.00		10.00	5.00	2.0		1		1	

Circle/Division	Preventive structures/measures required									
	Length in kilometers						Numbers			
	Elephant-proof Trenches	Elephant-proof Wall	Solar Power Fence	Hanging Fence	Bio-fence	Kayyala	Early Warning System	Radio-telemetry	Drone for scaring	Problem elephant relocation
NORTHERN CIRCLE KANNUR	**20.00**		**125.00**	**185.00**	**10.00**		**5**		**5**	**3**
Kannur			43.00	30.00	2.0		1		1	
Kasaragod	12.00		15.00	45.00	2.0				1	
Kozhikode	6.00		45.00	20.00	2.0		1		1	1
Wayanad North	2.00		20.00	55.00	2.00		2		1	1
Wayanad South			2.00	35.00	2.0		1		1	1
SOUTHERN CIRCLE KOLLAM	**30.00**		**150.00**	**85.00**	**15.00**		**5**		**6**	**1**
Achencoil	2.00		5.00	10.00			1		1	
Konni			45.00	10.00	10.00		1		2	
Punalur	2.00		15.00	25.00	1.00		2		1	
Ranni	1.00			20.00	4.00		1		1	
Thenmala	3.00		40.00	10.00						
Thiruvananthapuram	22.00		45.00	10.00					1	1
WILDLIFE CIRCLE PALAKKAD	**10.00**		**20.00**	**125.00**	**10.00**		**5**	**6**	**5**	**3**
Aralam	6.00		14.00	20.00	2.50		2	3	1	
Parambikulam										
Peechi	4.00			5.00	2.50		1		1	
Silent Valley				25.00					1	1
Wayanadu (WL)			6.00	75.00	5.00		2	3	2	2
Grand Total	**132.00**		**680.00**	**685.00**	**85.00**		**32**	**8**	**36**	**13**

Annexure 4

Table A4: Requirement of other activities to mitigate HWC in the Forest Divisions of Kerala

Circle/Division	Requirement of other activities to mitigate HWC									
	Vista clearance (Ha.)	RRT (No. of Divisions)	Rescue Centre (No.)	Elephant Squad (No.)	Create Waterholes (No.)	Maintenance Waterholes (No.)	Vayal maintenance (Ha.)	Weeds management (Ha.)	Equipment requirement (No. of Divisions	Training requirement (No.)
AGASTHYAVANAM BIOLOGICAL PARK	10				1	30		30	2	4
Shenthuruni					0				1	
Thiruvananthapuram WL	10				1	30		30	1	4
CENTRAL CIRCLE THRISSUR	234	1		1	48	55	17	1230	4	6
Chalakudy	25				20	15	7	600	1	
Malayattoor	150	1		1	15	10		50	1	5
Thrissur									1	1
Vazhachal	59				13	30	10	580	1	
EASTERN CIRCLE PALAKKAD	256	4		10	40	27	40	70	4	11
Mannarkkad	16	1			10	6			1	2
Nenmara	50	1			4					
Nilambur North	40	1		10	12	10	10	10	1	1
Nilambur South	120	1			6	11	30	60	1	4
Palakkad	30				8				1	4
FIELD DIRECTOR (PROJECT TIGER) KOTTAYAM	352		2			167	165	101	3	8
Idukki WL	22		1			25	25	1		
Munnar WL	30		1			25		100	1	6
Periyar East	150					64	100		1	1
Periyar West	150					53	40		1	1
HIGH RANGE CIRCLE KOTTAYAM	25	3	2	10	18	13	9	323	5	8
Kothamangalam		1							1	
Kottayam	18	1	1		16	5	9	63	1	4
Mankulam	7			1		2		110	1	
Marayoor	0			8	2	6		50	1	2
Munnar		1	1	1				100	1	2
NORTHERN CIRCLE KANNUR	151	3	3		29	24	37	131	4	18

Circle/Division	Requirement of other activities to mitigate HWC									
	Vista clearance (Ha.)	RRT (No. of Divisions)	Rescue Centre (No.)	Elephant Squad (No.)	Create Waterholes (No.)	Maintenance Waterholes (No.)	Vayal maintenance (Ha.)	Weeds management (Ha.)	Equipment requirement (No. of Divisions	Training requirement (No.)
Kannur	68	1	1		5				1	6
Kasaragod		1								
Kozhikode	50		1		10		10	100	1	4
Wayanad North	32				4	4	27	30	1	4
Wayanad South	1	1	1		10	20		1	1	4
SOUTHERN CIRCLE KOLLAM	230	2	3	25	51	142	10	160	6	24
Achencoil	50		1		5	5			1	2
Konni	75	1			10	50		100	1	10
Punalur		1	1	16	9	22		50	1	2
Ranni	30		1	9	9	13	10		1	4
Thenmala	40				6	6			1	
Thiruvananthapuram	35				12	46		10	1	6
WILDLIFE CIRCLE PALAKKAD	905			3	10	265	1010	240	3	10
Aralam	10			3		15		10		3
Parambikulam										
Peechi	50				10	15		50	1	5
Silent Valley	45						80	80	1	2
Wayanadu (WL)	800					235	930	100	1	
Grand Total	**2163**	**13**	**10**	**49**	**197**	**723**	**1288**	**2284**	**31**	**89**

Note: Please note that the division-wise figures under various items in this Annexure may not tally with the Circle-level figures given in the Tables provided along with running text, as the circle-level figures were changed during further discussions with field staff, higher officers and stakeholders on different occasions.

Annexure 5

Table A5: Presence of Rapid Response Teams in the forest divisions of Kerala

Circle	#	Division	Presence of RRT (Yes/No)	Location	Status
Southern Circle, Kollam	1	Thiruvananthapuram	Yes	Kuttichal	Sanctioned
	2	Thenmala	Yes	Anchal Range	
	3	Achencoil	No	-	
	4	Ranni	Yes	Ranni	Sanctioned
	5	Punalur	Yes	Anchal Range HQ	Sanctioned
	6	Konni	Yes	Konni	
High Range Circle, Kottayam	7	Kothamangalam	No	-	
	8	Munnar	Yes	Devikulam	
	9	Marayoor	No	-	
	10	Mankulam	No	-	
	11	Kottayam	No	-	
Central Circle, Thrissur	12	Vazhachal	No	-	
	13	Chalakudy	No	-	
	14	Malayattoor	No	-	
	15	Thrissur	No	-	
Eastern Circle, Palakkad	16	Mannarkkad	Yes	1. Mannarkkad RRT-Mannarkkad 2. Agali RRT-Sholayur	
	17	Nilambur North	Yes	Aruvacode, Nilambur	Sanctioned
	18	Nilambur South	No	-	
	19	Palakkad	Yes	Kallekulangara	Sanctioned
	20	Nenmara	No	-	
Northern Circle, Kannur	21	Kozhikode	Yes	Thamarassery	Sanctioned
	22	Wayanad North	No	-	
	23	Wayanad South	No	-	
	24	Kannur	Yes	Aralam farm, Aralam wild life Division	Sanctioned
	25	Kasaragod	Yes	Karaduka	
ABP Circle, Thiruvananthapuram	26	Thiruvananthapuram WL	No	-	
	27	Shenthuruni	No	-	
Field Director (PT), Kottayam	28	Periyar East	No	-	
	29	Periyar West	No	-	
	30	Munnar WL	No	-	
	31	Idukki WL	No	-	
Wildlife Circle, Palakkad	32	Parambikulam	-	-	
	33	Wayanadu (WL)	Yes	Sulthan Battery	Sanctioned
	34	Silent Valley	No	-	
	35	Peechi	-	-	
	36	Aralam	Yes	10th Block	

Annexure 6

Table A6: Circle Wise Details of existing Jana Jagratha Samithis

Sl No.	Name of Circle	No. of Jana Jagratha Samithies
1.	Sothern Circle, Kollam	32
2.	High Range Circle, Kottayam	43
3.	Central Circle, Thrissur	20
4.	Eastern Circle, Palakkad	46
5.	Northern Circle, Kannur	40
6.	Augasthyavanam Biological Park Circle, Thiruvananthapuram	5
7.	Wildlife Circle, Kottayam	10
8.	Wildlife Circle, Palakkad	8
	Total	204

Annexure 7

Table A7: Ex-gratia for Death and Injury due to Wildlife Attack

Sl No.	Details of loss to life and property / injury	Amount of Ex-gratia
1.	Death	Rs. 10.00 lakhs
2.	Death due to snake bite outside forest	Rs. 2.00 lakh
3.	Permanent disability	Maximum Rs. 2.00 lakh
4.	Injury	Maximum Rs. 1.00 lakh (People belonging to SC/ST community will be provided full expense for the treatment based on the medical certificate of a Medical Officer)
5.	Crop loss, cattle loss, damage to house	100% of the loss (Limited to Rs. 1.00 lakh for each incidence and 4 times in a year)

Annexure 8

Table A8: Rate of ex-gratia payment for Crop Damage due to Wildlife Attack

Name of crop	Rate of ex-gratia to be payable for loss of crops (Rs.)
1. Paddy (per ha)	11,000
2. Coconut (bearing per No)	770
3. Coconut (non- bearing per No)	385
4. Coconut (upto 1 year per No)	110
5. Banana (bunched per No.)	110
6. Banana (non- bunched per No.)	83
7. Rubber (tapping per No.)	330
8. Rubber (non-tapping per No.)	220
9. Cashew (bearing per No.)	165
10. Cashew (non-bearing above 3 years per No.)	110
11. Arecanut (bearing per No.)	165
12. Arecanut (non-bearing per No.)	110
13. Cocoa (bearing per No.)	110
14. Coffee (per No.)	110
15. Pepper (bearing per No.)	83
16. Ginger (for 10 cent)	165
17. Turmeric (for 10 cents)	132
18. Tapioca (Grown above two months)	165
19. Ground nut (per ha)	2200
20. Sesamum (for 50 cents)	1320
21. Vegetables (for 10 cents)	220
22. Nut meg (bearing per No.)	440
23. Nut meg (non-bearing per No.)	165
24. Clove (bearing per No)	220
25. Clove (non-bearing per No)	110
26. Cardamon (per ha)	2,750
27. Betel vine (1 cent)	330
28. Pulses (per 1 ha)	1,100
29. Tuber Crops (Colossiayam etc. for 10 cents)	165
30. Sugar cane (per ha)	2,750
31. Pineapple (for 10 cents)	825
32. Fodder grass (for 10 cents)	165
33. Mulberry (for 50 cents)	825
34. Tobacco (for 10 cents)	1650
35. Cotton (for 10 cents)	330

Annexure 9

Table A9: Identified Enclosures Inside Forests for Relocation

Sl. No	Division	Name of Settlement	Extent (Ha)	Total No. of families	Eligible families
1.	Shendurney	Rosemala, Kattilappara	110.75	237	398
2.	Marayoor	Puthuvettu, Edakkadu SC Colony, Edakkadu Colony, Karadi china para, Punavayal, Kundakkadacu	78.72	92	140
3.	Wayanad North	Pancharakolly, Maniyankunnu, Kalliyottukunnu, CRP Kunnu, Muriyanthadam	11.87	52	80
4.	Kasargode Forest Divn	Karveri, Karadukka, Kottamkuzhy, Adoor Muchaamthully, Ottamala, Thumbadukka, Kamballur, Vallakkadu, Balavathadukka	17.03	38	59

Annexure 10

Table A 10 Details of Identified Elephant Corridor

Sl No	Name of the corridor	Division under which the corridor falls	Corridor		Status of notification	Remarks
			Length in km	Width in km		
1.	Kottiyoor- Periya	Kannur, Wayanad (N)	3	0.1	Not notified	Land acquisition in progress
2.	Begur- Brahmagiri	Wayanad WLS with Brahmagiri WLS & Nagarhole TR (Karnataka)	1	0-0.8	Not notified	Land to be acquired
3.	Nilambur – Appankappu	Nilambur North & South Divisions	0.4	0.5	Not notified	Land to be acquired
4.	Periya at Pakranthalam	Wayanad (N)	0.5	0.2	Not notified	Land to be acquired
5.	Thirunelli- Kudrakote	Wayanad (N)	6	1 to 1.5	Yes	9.63 acres notifies as extension to Wayanad Wildlife sanctuary & remaining 8.37 acres as Reserve Forest.
6.	Nilambur Kovilakam-New Amarambalam	Nilambur North & South Divisions	1	0.5	Not notified	No acquisition needed
7.	Mudumalai – Nilambur via O' Valley (Ouchterlony Valley)	Nilambur North Forest Division and Mudumalai Tiger Reserve (TN)	35	0.1	Not notified	No acquisition needed on the Kerala side
8.	Shanamangalam – Padiri	Wayanad (N)			Not notified	Studies to be carried out
9.	Aryankavu	Thenmala Division			Not notified	Studies to be carried out

APPENDIX-1

PROCEEDINGS OF THE MEMBER SECRETARY

STATE PLANNING BOARD

(Present: Sri. Teeka Ram Meena IAS)

Sub: - Formulation of Fourteenth Five Year Plan (2022-27) – Constitution of Working Group on Forest and Environment – Revised Proceedings - reg.

Read: 1. Note No. 297/2021/PCD/SPB dated: 27/08/2021
2. Guidelines on Working Groups
3. This Office order of even number dated 08.09.2021

ORDER No:SPB/342/2021-Agri (4) Dated:14.09.2021

As part of the formulation of Fourteenth Five Year Plan, it has been decided to constitute various Working Groups under the priority sectors. Accordingly, the Working Group on **Forest and Environment sector** is constituted. For the smooth functioning of the Sectoral Working Group (SWG), it is decided to split the Working Groups into Expert Sub Groups (ESG). Hence the Working Group is categorized into four Expert Sub Groups as indicated in the proceedings. The names of the members of the SWG are indicated under each ESG. The Working Group shall also take into consideration the guidelines read 2nd above in fulfilling the tasks outlined in the ToR for the Working Group.

1. ENVIRONMENT AND BIODIVERSITY

Co - Chairperson

1. Dr T. Jayaraman, Former Member, Kerala State Planning Board
2. Suneel Pamidi ,IFS, Director, Directorate of Environment and Climate Change

Members

1. Dr C. George Thomas, Chairman, Kerala State Biodiversity Board
2. Dr Raman Sukumar, Eminent Ecologist, Expert in Human Animal Interface
3. Dr A. B. Anitha, Former Executive Director, CWRDM, Kozhikode
4. Dr A. G. Pandurangan, Former Director, TBGRI, Palode
5. Dr John Mathai, Scientist-G (Rtd.), NCESS
6. Dr P. S. Harikumar, CWRDM
7. Dr Syam Viswanath, Director, KFRI
8. Dr K. V. Thomas (Retd NCESS)
9. Dr R. Prakashkumar, Director, JNTBGRI
10. Mr B. Pradeep Kumar, Chairman, Kerala Pollution Control Board (KPCB)
11. Mr Reney R. Pillai, Member Secretary, Kerala State Biodiversity Board
12. Dr A. Gopalakrishnan, Director, Central Marine Fisheries Research Institute, Kochi
13. Dr K. Ranjeet, Associate Professor, KUFOS
14. Mr Joseph Vijayan, Social worker & Activist

15. Mr C. Baldwin, Kerala Karshaka Sangham, Aryardram, Kundara, Kollam
16. Dr.Aravindan Nagarajan, Sr. Lecturer, Azim Premji University, Bengaluru

Terms of Reference

1. To briefly assess the available information on the impacts of climate change on Kerala and how public policy should begin to address them.
2. To suggest a comprehensive framework for the design and implementation of schemes related to environment and climate change in Kerala.
3. The framework suggested may outline special problems pertaining to specific sectors of concern including infrastructure, the likely impact of climate change in specific sectors like agriculture and the impact of sea-level rise and other climate impacts on coastal regions and ecosystems.
4. The framework suggested may also list specific measures and schemes for knowledge generation and/or specific action as appropriate in these sectors.
5. To assess the design and implementation of existing schemes of Government of Kerala under the Department of Environment and Climate Change and suggest measures to improve their scientific orientation.
6. To outline special problems pertaining to biodiversity conservation, suggest a road map for their development, and examine the role of various agencies/departments/local self-governments in the implementation of biodiversity conservation-related programmes.
7. To suggest measures to better converge the activities and programmes of different line departments and agencies to meet the challenges of climate change and preservation of biodiversity.

2. ADDRESSING ISSUES RELATED TO HUMAN-WILDLIFE INTERACTIONS

Co - Chairperson

1. Mr P. K. Kesavan IFS, Principal Chief Conservator of Forests & Head of Forest Force
2. Shri. Bennichen Thomas IFS, PCCF(Wildlife)& Chief Wildlife Warden

Members

1. Dr P. S. Easa, Department of Wildlife
2. Dr Mani Chellappan, Professor, College of Horticulture, Vellanikkara
3. Mr O. P. Kaler, (Retd) APCCF, Kerala
4. Mr James Zacharia, (Retd) Forest Officer, Kerala
5. Dr S. Nandakumar, SIAD, Palode
6. Mr P. Viswan, Kerala Karshaka Sangham, Palothari, Koyilandi, Kozhikode
7. Mr K. Ramachandran, Adivasi Kshema Samithi, Panamaram

Terms of Reference

1. To assess the various interventions of the Government of Kerala in addressing the issues arising from human-wildlife interactions over the past decade.

2. To document the economic cost of crop losses arising out of human-wildlife interactions over the past decade.
3. To document the human and animal losses arising out of human-wildlife interactions over the past decade.
4. To suggest a multi-pronged plan of action on the different measures to be adopted to reduce human and animal losses as well as crop damages arising out of human-wildlife interactions over the next 10 years.
5. To ensure that the suggested plan of action includes a mixture of traditional and modern measures and that context-specific solutions are prioritised over generalised blanket solutions.
6. To suggest changes in forest governance and administration to allow a more effective resolution of issues arising out of human-wildlife interactions.

3. AN ASSESSMENT OF THE IMPLEMENTATION OF THE FOREST RIGHTS ACT IN KERALA

Co - Chairperson

1. Dr Geetanjoy Sahu, Associate Professor, Tata Institute of Social Sciences, Mumbai
2. Mr D. Jayaprasad IFS, PCCF (Planning and Development), Kerala Forests and Wildlife Department

Members

1. Mr P. N. Unnikrishnan, (Rted) PCCF, Kerala Forest Department
2. Dr A. V. Santhoshkumar, Professor and Head, Department of Forest Biology and Tree Improvement, College of Forestry, Thrissur
3. Dr G. Shine, Assistant Professor, College of Forestry, Thrissur
4. Mr O. R. Kelu, MLA, Mananthavady
5. Dr V. R. Najeeb, Independent researcher, Wayanad
6. Mr P. K. Suresh, Kerala Karshaka Sangham, Sarayu, Kammana, Wayanad
7. Mr.G.Anilkumar,Deputy Director,Tribal Resettlement &Development Mission,Vikas Bhavan
8. Mr.C.Herald John,Tribal Developmemnt Officer,Parappa,Kasargod.

Terms of Reference

1. To undertake a detailed assessment of the implementation of Forest Rights Act in Kerala between 2006 and 2021.
2. To identify gaps in policy and suggest measures to improve the implementation of the Forest Rights Act in Kerala over the next five years.
3. To identify outstanding issues in the legal structure and administrative reforms to facilitate a smoother implementation of the Forest Rights Act in Kerala.
4. To suggest a framework for a regular monitoring of the implementation of the Forest Rights Act in Kerala over the next five years.

4. TOWARDS A SUSTAINABLE MANAGEMENT OF FORESTS IN KERALA

Co - Chairperson

1. Prof T. K. Kunhamu, Professor, College of Forestry, KAU
2. Dr A. V. Raghu, Scientist, KFRI

Members

1. Mr S. Santhosh Kumar, ACF (Planning), Kerala Forests and Wildlife Department
2. Mr Tiju C. Thomas, Associate Coordinator-Communities, WWF India
3. Dr P. Niyas, Assistant Professor, College of Forestry, Thrissur
4. Ms T. R. Suma, Scientist, Hume Centre for Ecology and Wildlife Biology, Kalpetta
5. Mr Balan Madhavan, Senior Fellow, International League of Conservation Photographers
6. Adv. K. J. Joseph, Kerala Karshaka Sangham, Kuryasseri House, Chunkakunnu, Kottiyoor
7. Mr Rajith M R, Assistant Professor, Department of Economics, Dr.B.R Ambedkar Memorial Arts and Science College.

Terms of Reference

1. To critically assess efforts made in the last decade to improve the sustainability of forest management in Kerala.
2. To identify gaps in policy and suggest measures for improvement over the next five years.
3. To document the new global developments in the field of forest management and suggest best practices for adoption in Kerala.
4. To suggest measures to ensure that future policy on forest management blends the objectives of forest conservation and the livelihoods of people living inside and near forests.
5. To suggest measures to ensure that community participation is deepened in the design and implementation of the government's programmes and schemes towards forest management.

Convener

Mr.S.S.Nagesh, Chief, Agriculture Division, State Planning Board

Co- Convener

Dr.Reji D Nair, Research Officer, Agriculture Division, State Planning Board

Terms of Reference (General)

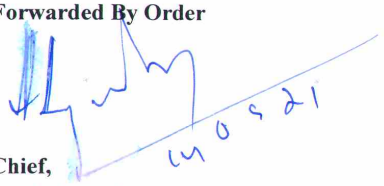
1. The non-official members (and invitees) of the Working Group will be entitled to travelling allowances as per existing government norms. The Class I Officers of GoI will be entitled to travelling allowances as per rules if reimbursement is not allowed from Departments.

2. The expenditure towards TA, DA and Honorarium will be met from the following Head of Account of the State Planning Board "3451-00-101-93"- Preparation of Plans and Conduct of Surveys and Studies.

The order read as reference 3 is modified to this extent.

(Sd/-)
Member Secretary

Forwarded By Order



Chief,
Agriculture Division

To

The Members concerned

Copy to

PS to Vice Chairperson
PA to Member Secretary
CA to Member (Dr.Ramakumar.R)
Economic Advisor to VC
Chief, PCD,SPB
Sr. A.O, SPB
The Accountant General, Kerala
Finance Officer, SPB
Publication Officer, SPB
Sub Treasury, Vellayambalam
Accounts Section
File/Stock File